Long-Tail Compositional Road-Event Detection:

A Decoupled Hybrid with Learned and Semantic Composition on ROAD-Waymo

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Abstract

Road-event detection identifies agents, actions, scene locations, and their compositions. In ROAD-Waymo, unequal training support makes rare compositions difficult to recognize. This thesis investigates how decoupled localization, learned composition, and language-informed contextual representations affect recognition of these events.

A YOLOv8x detector supplies candidate boxes and agent identities, while separately trained event heads use frozen video features. The study progresses from overlap-based score transfer and direct region classification to crop-based semantic classification, scene and language attention, and complementary expert blending. The final Contextual Semantic Expert Blend combines a dynamic-class-balancing classification expert with a second expert trained using an additional visual–phrase contrastive objective. A held-out training partition selects one fixed probability-blending weight per seed. Larger paired video and text encoders supply the final system's frozen representations.

Evaluation uses frame-level mean average precision (mAP) at IoU 0.5 on 36,717 shared ROAD-Waymo validation frames. Three-seed means reach 15.75% triplet mAP and 9.00% tail mAP over the 47 classes below the geometric mean training frequency, versus 7.54% and 4.14% for the replicated joint baseline. Scaling increases triplet mAP by 2.32 points over the original-encoder DCB blend; about 65% of this increase comes from tail classes, while common-class performance also improves. Controlled comparisons show that benefits depend on language, context, and loss: phrase-score fusion alone produces small changes, while DCB with original encoders improves overall performance but reduces tail performance. These findings support complementary contextual representations and expert predictions, with limitations from validation reuse, fixed-vocabulary evaluation, and system-level changes in the scaling comparison.

Biographical Sketch

Brandon L. Byrd holds an Associate of Applied Science in Aircraft Maintenance Technology from the Community College of the Air Force and a Bachelor of Science in Computer Science from North Carolina Agricultural and Technical State University. He is pursuing a Master of Science in Computer Science at North Carolina Agricultural and Technical State University under the supervision of Dr. Hamidreza Moradi. His research focuses on autonomous-driving perception and compositional road-event detection. His thesis investigates a decoupled detection and classification pipeline and the use of semantic structure for rare event categories.

Dedication

In memory of Professor Leflore, whose Python 163 course challenged me deeply and helped shape my growth as a computer scientist.

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CHAPTER 1: Introduction

Scene understanding supports the broader goal of safer driving by describing the road users and events around a vehicle. Multiple agents act at once, visibility changes with ego motion and occlusion, and an action may require temporal context to interpret. This motivates a perception system that connects an object with its observed behavior and road location. The present thesis addresses that recognition problem; the evaluation measures detection performance rather than driving decisions or collision risk.

Autonomous-driving perception must describe more than which objects are visible. Road-event understanding also concerns what an agent is doing and where that action occurs. ROAD and ROAD-Waymo express these relationships through agent, action, and location labels and their compositions [1], [2]. A detector that recognizes a cyclist can still fail to recognize the cyclist's action in the relevant road location.

This thesis studies long-tail compositional road-event detection on ROAD-Waymo. The task predicts boxes and scores for agentness, agent identity, actions, locations, duplexes, and triplets. A duplex combines agent and action; a triplet adds location. These are labels for observed events in a contextual clip, rather than direct measurements of mental intent.

Figure 1 makes the prediction target concrete. The highlighted car has the primitive labels Car, Stop, and VehLane, meaning that it is stopped in the ego vehicle's lane. Their duplex is Car-Stop and their triplet is Car-Stop-VehLane. Recognizing the car alone therefore answers only one part of the road-event task. The label describes an annotated observation, not the driver's future intention.

The difficulty is that composition categories receive highly unequal training support. Common events contribute many examples, while a rare combination may have few labeled boxes even

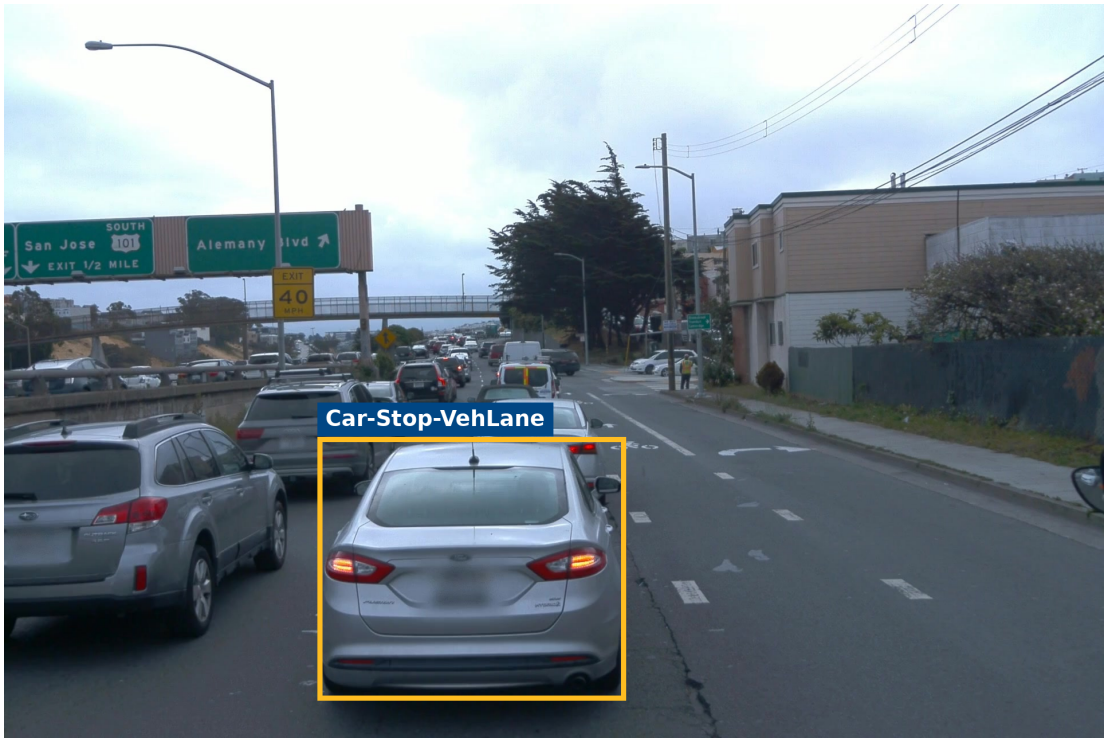


Figure 1: A labeled ROAD-Waymo road event [2]. One ground-truth box is shown from video train_00407, frame 00001, annotation b_49; other boxes are omitted for clarity.

when its constituent agent or action is familiar. An overall mean can therefore conceal persistent weaknesses in rare-event recognition. Improving localization is useful, but it does not establish that a classifier can recognize the event associated with each newly detected object.

1.1 Compositional Imbalance and the Research Problem

The local training-count analysis covers 86 triplet classes. Counts range from 61 to 225,813 boxes, a ratio of approximately 3,702 to one. For class c with training count n_c, define $l_c = \log_{10}(n_c)$, μ as the mean of those 86 log counts, and σ as their population standard deviation. The standardized value is $z_c = (l_c - \mu)/\sigma$.

The geometric mean training count is $10^{\mu} \approx 6{,}434$. The 47 classes with $z < 0$ form the principal tail group. A nested deeper-tail group contains the 28 classes with $z < -0.5$. The remain-

ing 39 classes form the complementary common group. Membership depends only on training counts and stays fixed across model comparisons.

These groups are analytical definitions for this study, not claims that rarity measures driving risk. A rare label is not automatically more hazardous than a common one. The motivation is to make unequal recognition performance visible and test a mechanism that may help categories with limited training support.

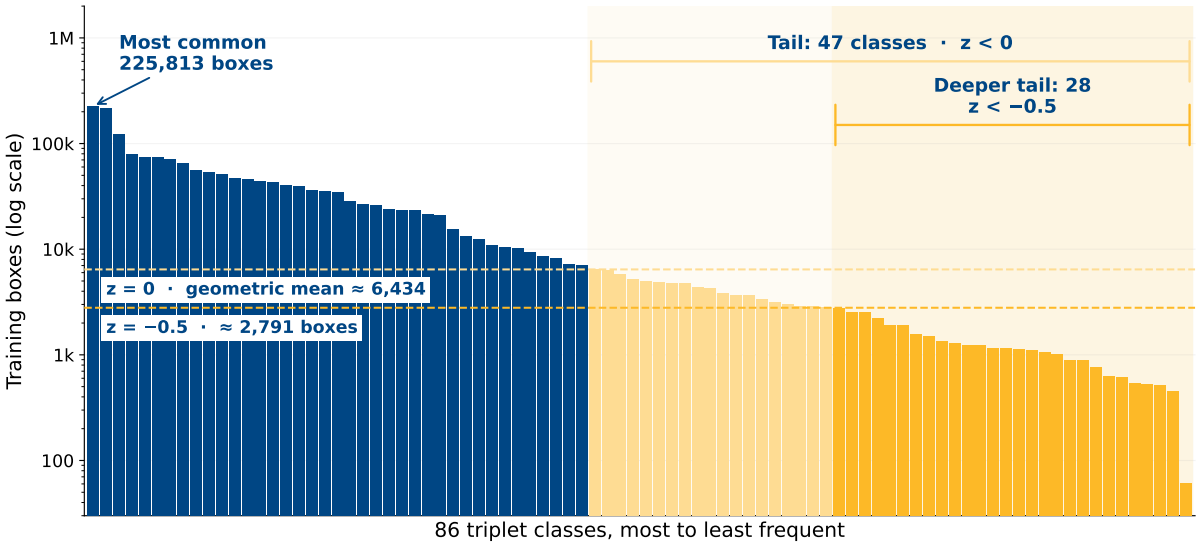


Figure 2: ROAD-Waymo triplet training counts and tail definitions.

Figure 2 shows the training-frequency imbalance and the nested evaluation groups. The z-score is computed on log counts: $z < 0$ is equivalent to a count below the geometric mean, while $z < -0.5$ identifies classes farther below it. These thresholds define evaluation populations; they do not assert that the counts follow a normal distribution.

1.2 Research Objectives and Developed Approach

The central question is whether rare road-event compositions benefit from shared structural and semantic information while retaining useful recognition of common classes. The study examines detector replacement, direct region-feature access, learned composition, semantic crop classification, and contextual fusion. Controlled comparisons ask whether gains accompany meaningful phrase directions, scene evidence, the classification loss, or complementary expert predictions.

The experimental sequence begins with a replicated 3D-RetinaNet baseline. Stage 1 transfers its event scores to YOLO boxes through overlap matching. Stage 2 extracts frozen I3D region features at every candidate box and trains a linear scoring head. Stage 3 replaces independent composition outputs with a composition MLP. These stages distinguish improvements in localization from improvements in event classification.

Stages 4–6 use frozen InternVideo2 crop descriptors. Stage 4 projects crop features toward fixed event-phrase directions. Stage 5 learns compositions from flat primitive scores and crop features; Stage 6 additionally supplies phrase-composition scores to that MLP. The subsequent contextual family combines crop, actor-region, entire-frame, geometry, and phrase evidence in a direct event classifier. Scene attention and visual–phrase contrastive learning provide distinct ways to use context and language. The final Contextual Semantic Expert Blend (CSEB) combines separately trained classification and contrastive experts, with dynamic class balancing and scaled frozen encoders.

1.3 Contributions and Scope

The thesis provides an explicit evaluation of the detector–classifier development sequence and its effects on rare compositions. The controlled Stage 2-to-3 comparison increases triplet f-mAP from 6.65% to 9.73% while retaining candidate boxes and primitive outputs. The final

CSEB reaches 15.75% triplet and 9.00% tail mAP as a three-seed mean. Comparisons within the contextual family show that improvements in overall mAP can coexist with tail regressions, and that blending complementary experts can improve both groups.

The methodological contribution is the integration and controlled assessment of these components for ROAD-Waymo, rather than a claim that attention, semantic classifiers, dynamic class balancing, or probability blending are individually new. The study reports all prediction groups, fixed training-count-defined tail subsets, and seed variability. External published comparisons retain their different detection and training protocols.

The scope is frame-level detection within a fixed ROAD-Waymo vocabulary. Context clips can include frames after the evaluated frame. The results therefore concern observed events, not causal intent prediction. Independent test-set confirmation, deployment safety, and unseen-composition generalization remain outside the demonstrated claims.

CHAPTER 2: Literature Review

This thesis draws on research in road-event detection, long-tailed recognition, video representation learning, and vision–language classification. These areas address different parts of the problem: detecting an agent, representing its behavior, learning from unequal class frequencies, and sharing information among related event descriptions. The review distinguishes the evidence established in each setting from the questions that remain for rare ROAD-Waymo compositions. In particular, improvements in image classification, video recognition, or unseen attribute–object recognition are motivations for this study, rather than direct evidence of improved road-event detection.

2.1 Road-Event Benchmarks and Visual Representations

Pedestrian-behavior research illustrates why detection alone is insufficient for scene understanding. JAAD pairs pedestrian boxes with behavioral and contextual annotations, enabling study of crossing behavior and interaction with traffic [3]. PIE adds human-referenced crossing-intention estimation and future trajectory prediction [4]. These tasks connect appearance, motion, and context, but their targets must be distinguished: an observed crossing action, a future crossing outcome, and a human judgment of intention are different labels. ROAD extends the event-recognition setting across road-agent categories through agent, action, and location compositions. This thesis adopts that event vocabulary and does not transfer pedestrian-intention interpretations to ROAD-Waymo action labels.

Driving datasets differ in the relationships their annotations support. The Waymo Open Dataset provides synchronized camera and LiDAR observations with object annotations for perception research across varied locations [5]. BDD100K supports heterogeneous driving tasks, including detection and other forms of scene understanding [6]. These benchmarks establish the

importance of localization and diverse observations, but object identity alone does not describe the event in which an agent participates. AVA offers a complementary action-detection setting: it associates spatially and temporally localized people with multiple atomic action labels [7]. Its relevance is the need to classify an actor within a scene rather than assign a single label to an entire video. Its movie-based action vocabulary and evaluation setting differ from those of road-event detection.

ROAD explicitly represents road events through agents, actions, locations, and their compositions. Singh et al. introduced the benchmark and a 3D-RetinaNet baseline for event detection [1]. Duplex labels combine agent and action, while triplets also include location. ROAD-Waymo applies compatible event annotations to US driving video from Waymo Open, allowing comparison with the original UK ROAD data within the ROAD++ framework [2]. The present thesis uses ROAD-Waymo rather than combining the two domains. Consequently, evidence about geographical transfer and evidence about rare compositions within ROAD-Waymo must remain separate.

ROAD-R adds explicit logical requirements to ROAD and studies learning and prediction mechanisms that use those requirements [8]. This distinguishes two meanings of structure. Logical requirements specify which combinations are admissible; semantic representations describe similarities between class meanings. Encouraging constraint satisfaction during training is also different from enforcing admissibility at prediction time. A classifier may produce logically compatible outputs while failing to recognize the correct event. Conversely, improved average precision does not guarantee consistency between primitive and composition predictions. The semantic classifier therefore addresses recognition and does not claim the guarantees of a constrained-output method.

The replicated detector connects several foundational visual architectures. Residual networks make deep representations easier to optimize through residual mappings [9]. Inflated 3D

networks extend image filters and pooling operations into the temporal dimension to learn video features [10]. Feature Pyramid Networks combine a top-down pathway with lateral connections to obtain representations at several spatial scales [11]. RetinaNet couples dense detection with focal loss, which reduces the training contribution of easy examples [12]. ROAD's 3D-RetinaNet adapts this detection approach to road events [1]. The local baseline uses a ResNet50-I3D backbone, so the I3D citation establishes the inflation principle without implying that the local model is the original paper's exact backbone.

Region features provide a basis for separating localization from classification. Mask R-CNN introduces RoIAlign to preserve spatial alignment when extracting a fixed-size representation for a region [13]. In the developed pipeline, a detector supplies the box and RoIAlign reads a frozen video feature map at that box. This separates whether an object was proposed from what features are available to classify it. It also clarifies why copying scores from an overlapping detection is a different operation: score transfer remains dependent on the original detector producing a suitable match.

Video representations provide further alternatives to a fixed backbone. SlowFast networks use a slow pathway for spatial semantics and a faster pathway for fine temporal information [14]. Video Swin Transformer introduces local spatiotemporal attention through a hierarchical transformer design [15]. VideoMAE learns video representations by reconstructing heavily masked video content, emphasizing self-supervised pretraining [16]. These approaches demonstrate that temporal sampling, architecture, and pretraining objectives can all influence the representation. They motivate controlling the visual input when evaluating semantic classifiers: replacing both an encoder and a classifier cannot isolate the effect of language-derived class structure.

2.2 Actor Context, Memory, and Dynamic Balancing

Multiscale Vision Transformers build a hierarchy with decreasing spatial resolution and increasing channel capacity [17]. MViTv2 adds decomposed relative positional embeddings and residual pooling connections [18]. MeMViT extends temporal support by attending to cached representations from earlier processing steps [19]. These mechanisms improve how video evidence is represented or retained. They differ from the present contextual head, which combines frozen crop, region, and scene descriptors within the sampled clip and additionally uses a phrase bank.

STMixer adaptively samples spatiotemporal features and mixes them along spatial and temporal dimensions in a sparse action detector [20]. CycleACR reorganizes actor-conditioned context and uses that context to enhance actor features [21]. Their original action-detection evaluations include AVA, whose human-action vocabulary differs from ROAD-Waymo events. ROAD-Waymo values reproduced later in FRCB are external evaluations of these model families, not results from their original AVA experiments. These methods motivate richer actor-context interaction but do not by themselves test this thesis's language pathway.

FRCB combines an MViT-based video representation with feature modulation, local and historical context, and dynamic class balancing for road-event detection [22]. Its loss uses current and accumulated class difficulty to weight binary classification errors. The present study adapts that loss principle to frozen-feature contextual heads with explicit implementation choices. FRCB's published ROAD-Waymo best-result triplet f-mAP is 11.71%, while its five-run mean is 11.53 $\pm$ 0.35%. Its evaluation does not provide this thesis's exact candidate manifest or fixed 47-class tail subset. The comparison therefore motivates the loss experiment and supplies published context without establishing matched-protocol superiority.

2.3 Learning Under Long-Tailed Class Distributions

Class imbalance can affect both the representation and the decision rule learned from it. Focal loss addresses the dominance of easy examples in dense detection [12]; this is related to, but not identical to, unequal support among foreground event categories. A rare triplet may remain difficult even after background examples receive less influence. Thus, the use of focal loss in the developed pipeline does not by itself establish that rare compositions have been adequately addressed.

Several approaches adjust how class frequency enters the learning objective. Cui et al. derive class-balanced weights from the effective number of samples, reflecting diminishing additional information as the number of examples grows [23]. Cao et al. introduce a label-distribution-aware margin and defer reweighting until after an initial representation-learning period [24]. Both methods use class frequency, but they intervene differently: one changes the contribution of examples, while the other changes class-dependent margins and the training schedule. For this study, these distinctions reinforce the need to define tail membership from training data and to keep that definition fixed during evaluation.

Balanced Meta-Softmax addresses distribution shift between training and evaluation through Balanced Softmax and a complementary sampling mechanism [25]. Menon et al. study frequency-based logit adjustment either after training or within the loss [26]. These methods show why a change in class scores may improve a class-balanced metric without changing the underlying features. Their softmax formulations also assume a multiclass setting that differs from the independent sigmoid outputs used for multi-label road events. Applying their conclusions directly to the semantic classifier would require an appropriate multi-label formulation and a separate experiment.

Decoupling offers an experimental way to study these effects. Kang et al. separate representation learning from classifier learning and show that adjusting a classifier over a naturally sampled representation can be effective on long-tailed image benchmarks [27]. BBN uses separate branches and a cumulative learning schedule to balance representation and classifier learning [28]. These studies motivate treating the classifier as an explicit component rather than attributing every change to the encoder. The developed road-event pipeline follows this principle through frozen features and separately trained heads, while recognizing that image-level classification and box-level event detection have different supervision and error sources.

Other methods change representation geometry or model diversity. RIDE uses multiple distribution-aware experts to address bias and variance under long-tailed sampling [29]. Parametric Contrastive Learning introduces learnable class centers to counter the tendency of supervised contrastive learning to favor frequent classes [30]. These approaches show that shared structure can be imposed through more than language. A semantic classifier should therefore be assessed against a meaningful classifier control, rather than compared only with a weak independent-label baseline. They also motivate reporting common-class behavior, since improvements for rare categories may be accompanied by regressions elsewhere.

VL-LTR is especially relevant because it introduces class-level text descriptions into long-tailed visual recognition [31]. It learns visual and linguistic representations and reports benefits for classes with fewer image examples. This provides evidence that language can contribute to recognition under imbalance, but it does not establish the same outcome for multi-label road-event boxes. The crop study differs in using a fixed benchmark vocabulary, frozen crop features, and fixed phrase embeddings with a supervised projection. Its matched controls test whether those design choices help rare triplets under the road-event protocol.

Taken together, these methods identify several possible explanations for a tail improvement: better visual features, changes in sampling or loss, classifier calibration, additional capacity, and information shared among classes. The thesis evaluates learned composition, semantic classifier directions, contextual fusion, and dynamic class balancing. These components provide different mechanisms for sharing information and changing the influence of training examples. Their contributions must be distinguished through matched comparisons rather than attributed collectively to language.

2.4 Vision–Language Adaptation for Video Recognition

CLIP learns aligned image and text representations through contrastive pretraining, allowing class descriptions to define classifier directions [32]. The resulting relationship between language and visual content is the basis for considering event phrases as more informative than unrelated class identifiers. However, a description that is linguistically plausible may not distinguish visually similar actions or locations. The relevant question is whether the learned alignment retains the distinctions needed for road-event recognition.

Prompt-learning methods demonstrate both the usefulness and sensitivity of text-based classification. CoOp learns context vectors while keeping the pretrained model fixed [33]. CoCoOp conditions part of the prompt on the input to improve generalization beyond the classes used for adaptation [34]. MaPLe couples prompts in the visual and language branches to adapt their representations together [35]. These methods make different parts of the model trainable. They therefore cannot be treated as equivalent to a classifier whose phrase matrix remains frozen. For this thesis, their main implication is that the choice and adaptation of class descriptions must be documented and controlled.

Adaptation can also use labeled visual examples directly. Tip-Adapter builds a key–value

cache from a few-shot training set and combines feature retrieval with the pretrained model's predictions [36]. It provides a useful contrast to prompt learning: downstream performance may improve through visual evidence without changing the textual description. This motivates keeping the training population and feature source fixed when comparing the semantic head with a free classifier. A difference between those heads can then be interpreted more narrowly than a comparison in which the representations and examples also change.

Transferring image–text models to video introduces temporal design choices. X-CLIP uses cross-frame attention and video-dependent prompting to adapt pretrained image–language representations to video recognition [37]. Ju et al. combine continuous prompts with lightweight temporal transformers over frame features [38]. ViFi-CLIP shows that fine-tuning and pooling frame-level CLIP features can provide a strong video baseline without requiring a complex new temporal module [39]. Together, these studies caution against attributing an improvement to semantic reasoning merely because a vision–language model is involved. Temporal processing, adaptation, and pooling can each contribute.

InternVideo2 develops a family of video models through progressive training that combines masked video modeling, cross-modal contrastive learning, and generative objectives [40]. The original crop implementation uses the released InternVideo2_CLIP_S model, whose repository supplies its configuration and weights [41]. The family-level paper motivates the representation, while the model release identifies the specific component used in this thesis. The crop head reads an internal crop descriptor and learns a projection into the phrase space; it does not assume that the internal descriptor is already the released model's final video–text embedding.

The distinction between pretraining and the downstream experiment is central. Contrastive video–text learning has already shaped the frozen encoders. The Stage 4 ROAD-Waymo experi-

ment trains a supervised classifier using those representations. It tests whether retaining semantic directions is useful for rare benchmark compositions. The later contextual study adds a separate supervised visual–phrase contrastive objective at the head level while keeping the encoders frozen. Neither procedure reruns encoder pretraining, and the presence of text alone does not establish temporal reasoning.

2.5 Compositional Structure and the Research Gap

Compositional recognition asks how predictions for parts relate to predictions for their combinations. ROAD considers both direct composite prediction and composition of primitive scores [1]. The developed MLP follows the former goal through a structured input: it receives primitive predictions together with visual features and learns scores for the benchmark's valid duplexes and triplets. This gives it access to relationships that a fixed product or minimum of primitive scores cannot adapt to the task. It does not impose logical agreement between independently predicted heads.

Compositional zero-shot learning provides related evidence about sharing information among concepts. Naeem et al. connect states, objects, and their compositions through a graph so that relationships among primitives inform composition classifiers [42]. Nayak et al. learn compositional soft prompts in which attribute and object tokens can be recombined to describe unseen combinations [43]. These methods explicitly target generalization to compositions absent during training. The ROAD-Waymo study instead evaluates a fixed vocabulary that includes rare but observed classes. Their results motivate structured sharing, but do not justify a zero-shot claim for this thesis.

The literature therefore supports a focused gap rather than the claim that semantic classification is new in general. Long-tailed recognition studies frequency-sensitive losses, decoupled

classifiers, and linguistic information; video adaptation studies the transfer of image–text representations; compositional learning studies shared primitives and combinations. The empirical question is how phrase-based classifiers and language-informed contextual heads affect rare ROAD-Waymo triplets, and which comparisons isolate their contribution from representation, loss, or classifier changes.

The experiments use this platform to compare overall, common-class, and tail AP across repeated seeds. Matched controls qualify semantic and contextual explanations, including cases where additional structure gives mixed results.

CHAPTER 3: Methodology

We developed the ROAD-Waymo pipeline through controlled changes to a replicated joint detector. Detector replacement and score transfer are followed by direct region-feature classification and learned composition. This chapter defines the evaluation protocol and the completed methods, including semantic crop classification, contextual fusion, dynamic class balancing, and the scaled expert blend.

The completed stages share the benchmark label layout and evaluator. Changes in boxes, representation, classifier, and composition are described explicitly so that improvements are attributed only to the intervention supported by a comparison.

3.1 What Was Retained and What Was Modified

The starting point is the published ROAD detection framework and its 3D-RetinaNet video detector, evaluated on ROAD-Waymo [1], [2]. This baseline jointly predicts localization and event scores from an I3D feature pyramid. We retain the benchmark labels and evaluation code so that subsequent systems answer the same recognition task. The final model is not the baseline with one extra layer: development first separates localization from recognition, then replaces the event representation and readout.

Three sources of components must be distinguished. The ROAD detector supplies the replicated baseline and the frozen I3D features used in Stages 2 and 3. YOLOv8x supplies the later candidate boxes and agent outputs after ROAD-Waymo fine-tuning [44]. Released InternVideo2 paired encoders supply the frozen visual and phrase representations used by the crop and contextual families [40], [41]. We do not claim to have invented these backbones or rerun their original large-scale pretraining. The implemented contribution is the downstream composition and contextual recognition system, its training and score-assembly procedure, and the controlled comparison of

these alternatives.

Table 1 separates the retained components from each intervention. Stages 4 and 5 are alternative heads on the crop representation: Stage 5 does not consume Stage 4 predictions. Stage 6 introduces an explicit phrase-score branch. The contextual model then replaces the crop composition MLP with a direct 184-output classifier; it is not a seventh layer stacked on the earlier MLP.

Table 1: Changes relative to the preceding architectural reference.

Model	Retained	Changed and purpose
Stage 1	Baseline event detections	YOLO boxes replace baseline boxes; overlap transfers event scores to the new locations.
Stage 2	YOLO boxes; frozen I3D backbone	RoIAlign plus a trained linear head supplies event scores at every candidate, including unmatched boxes.
Stage 3	Stage 2 features and primitive outputs	A trained MLP replaces duplex/triplet outputs using primitive scores and visual evidence.
Stages 4–6	YOLO localization and label order	Frozen InternVideo2 crop features replace I3D RoI features; phrase, visual-composition, and phrase-fusion heads test different uses of semantic evidence.
Contextual	Frozen paired encoders; YOLO boxes	Actor-region, scene, geometry, and crop evidence are fused before language attention and direct event classification.
CSEB	Contextual-head architecture	Two independently trained experts are combined with a fixed selected probability weight; DCB changes training and paired-encoder scaling changes the feature source.

Figure 3 compares the starting and final information paths. In the baseline, one video representation serves box regression and event classification. In CSEB, YOLO owns localization and agent identity. Frozen video features describe the candidate crop, its region in the full frame, and the wider scene; frozen phrase embeddings provide semantic references. Trainable projections,

adapters, attention modules, and classifiers connect those representations. Only the composition probabilities are blended. Detector confidence is applied once at final emission, rather than inside the MLP or before expert mixing.

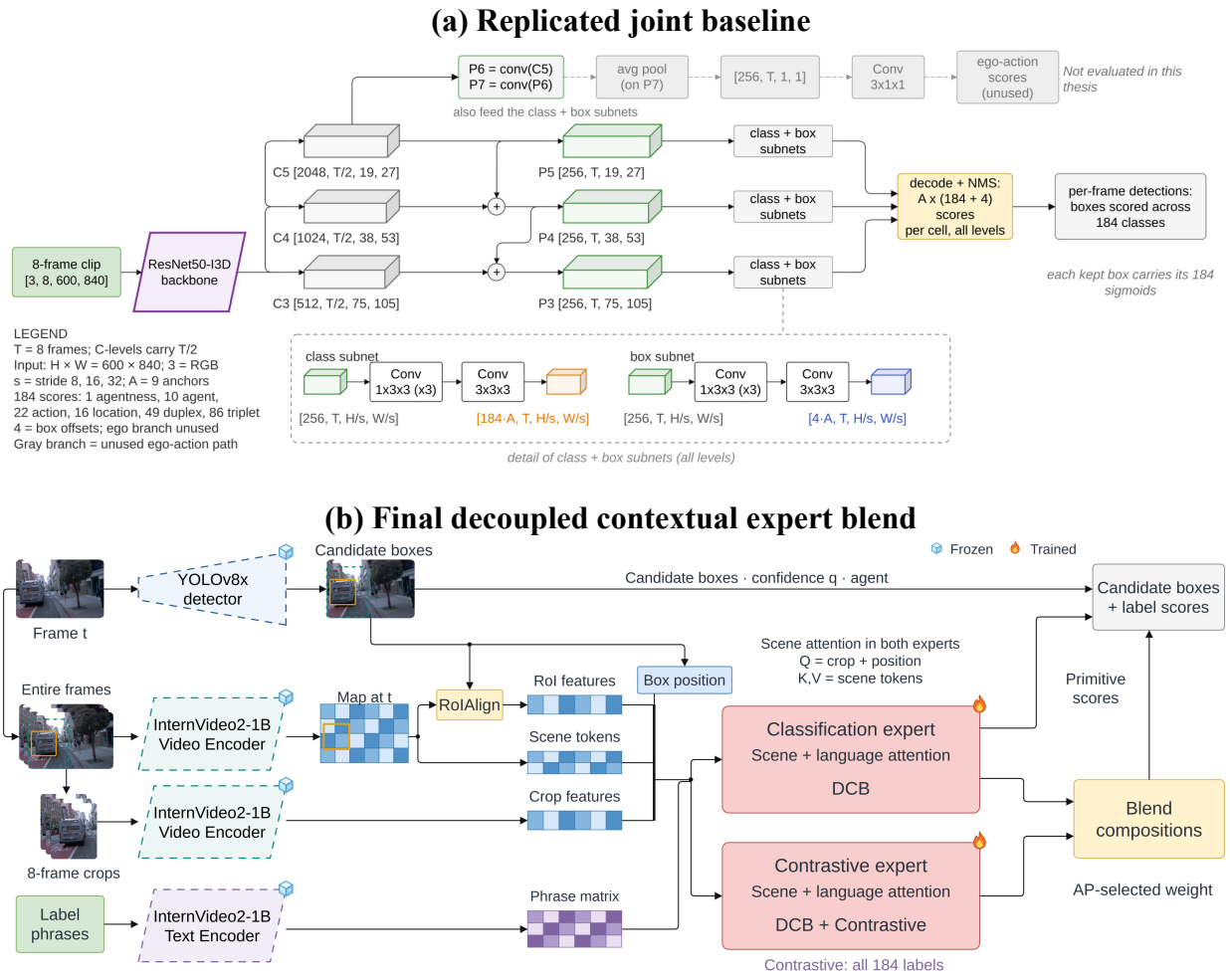


Figure 3: Baseline and final-model information paths. Existing approved architecture figures are shown together to distinguish joint box/event prediction from frozen feature extraction and separately trained contextual recognition. Backbone architectures are inherited; the thesis evaluates their downstream integration.

During downstream training, the final visual and text encoders are frozen and their outputs are cached. Each expert learns its own adapters, projections, attention, and classifier. The clas-

sification expert minimizes the classification objective; the contrastive expert additionally aligns pre-language visual features with the adapted phrase bank. After checkpoint selection, both experts are frozen and one scalar blend weight is selected on reserved training videos. At inference, neither ground-truth labels nor per-example optimization is used. These phases explain why a trained-module badge in a diagram does not imply end-to-end training of the pretrained encoders.

3.2 Dataset, Prediction Targets, and Evaluation

The study uses the ROAD-Waymo training and validation partitions [2]: 600 training videos and 198 validation videos. The publicly unannotated test partition is outside the reported evaluation. Dataset counts and tail membership refer to the local annotation release and project audit, rather than substituting aggregate counts reported for other releases.

Each candidate is associated with the score groups in Table 2. Agentness is a separate object-presence score. Action and location are multi-label tasks, and each composition is a benchmark-defined combination. The total is 1 + 10 + 22 + 16 + 49 + 86 = 184.

Table 2: Prediction groups and output dimensions.

Prediction groups	Dimensions	Meaning
Agentness + agent	1 + 10	Presence + identity
Action + location	22 + 16	Activity + position
Duplex + triplet	49 + 86	Event compositions
Total	184	Six output groups

All reported detector comparisons use frame-level mean average precision at IoU 0.5 through the baseline repository's evaluator. Predictions are ranked by class and matched to ground truth on each frame. AP summarizes the precision–recall curve; mAP averages class APs. The reported comparison is restricted to 36,717 validation frames shared by the detection sources and required feature records.

Figure 4 shows how a scored box contributes to this metric. For each class, the evaluator processes detections in descending score order within a frame and matches each detection to an available ground-truth box of that class when IoU is at least 0.5. Each ground-truth box can be matched once. Unmatched detections, including duplicate detections after a ground-truth box has been consumed, count as false positives. The evaluator then ranks detections across the selected frames to obtain a precision–recall curve. It integrates the interpolated precision envelope at changes in recall and averages AP over the classes in the reported group. This follows the detection-evaluation principles of PASCAL VOC [45]; the actual implementation is the ROAD baseline evaluator [1].

For boxes B and G, the matching criterion uses

$$\mathrm{IoU}(B, G) = \frac{|B \cap G|}{|B \cup G|} \geq 0.5. \tag{3.1}$$

Let R_k and P_k be recall and precision after the first k ranked detections, and let $\widehat{P}_k = \max_{j \geq k} P_j$ be the interpolated precision envelope. With $R_0 = 0$, the reported group metric is

$$\mathrm{AP}(c) = \sum_{k=1}^{n_c} (R_k - R_{k-1}) \widehat{P}_k, \qquad \mathrm{mAP}(\mathscr{C}) = \frac{1}{|\mathscr{C}|} \sum_{c \in \mathscr{C}} \mathrm{AP}(c). \tag{3.2}$$

Here n_c is the number of ranked detections for class c, and $\mathscr{C}$ is the reported class group. Multiplication by 100 expresses AP and mAP as percentages. The fixed tail and deeper-tail groups contain 47 and 28 triplet classes, respectively.

“Full-candidate” denotes a broad low-threshold candidate pool, capped at 300 per frame, rather than exhaustive boxes or every annotated validation frame. Stage 0 uses its own candidates. Stages 1–6 and the contextual models use the fixed YOLO candidate source. The original base-

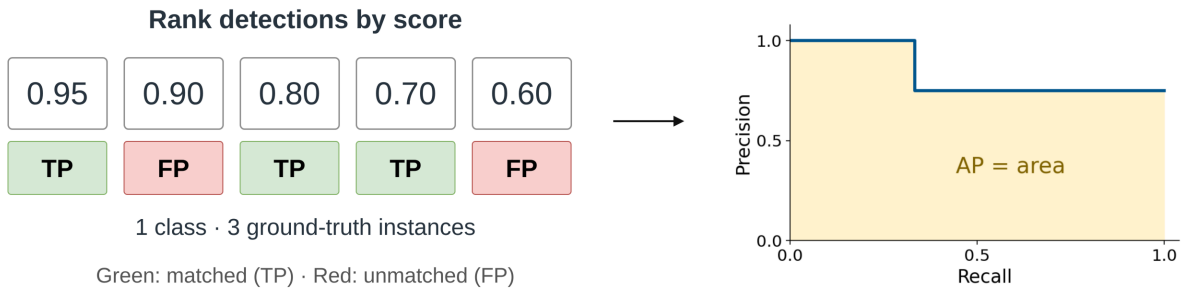


Figure 4: Frame-level average-precision evaluation. The slide example ranks five detections for a class with three ground-truth instances; green and red denote true and false positives. The precision envelope is schematic.

line validation path filters scores per class and per frame; the comparison path assembles predictions from saved dumps, including temporally averaged agentness for baseline candidate selection. Thus, using the same metric code alone does not make different filtering protocols equivalent.

The primary metric is f-mAP, not tube-level video-mAP. No test-server result is claimed. Training labels define tail membership, while validation predictions determine AP. The same tail class lists are used for every compared model.

3.3 Replicated Joint Baseline

The starting point is 3D-RetinaNet, which processes eight-frame clips through an I3D backbone and feature pyramid before predicting box offsets and class scores [1]. Localization and classification share a learned representation. The replicated checkpoint provides both the baseline detections and the frozen visual backbone used in the early hybrid stages.

Two evaluations are distinguished. The earlier baseline validation produced 17.76% agent and 9.17% triplet f-mAP. The Stage 0 row in this thesis's shared-frame comparison is 16.67% agent and 7.54% triplet f-mAP. The difference reflects evaluation conditions, including frame selection and candidate filtering; the numbers are not interchangeable.

Figure 5 details the ResNet50-I3D feature pyramid and detection subnets in the replicated 3D-RetinaNet. RetinaNet introduced focal loss for dense one-stage detection [12]; ROAD adapts this architecture to video road-event detection [1].

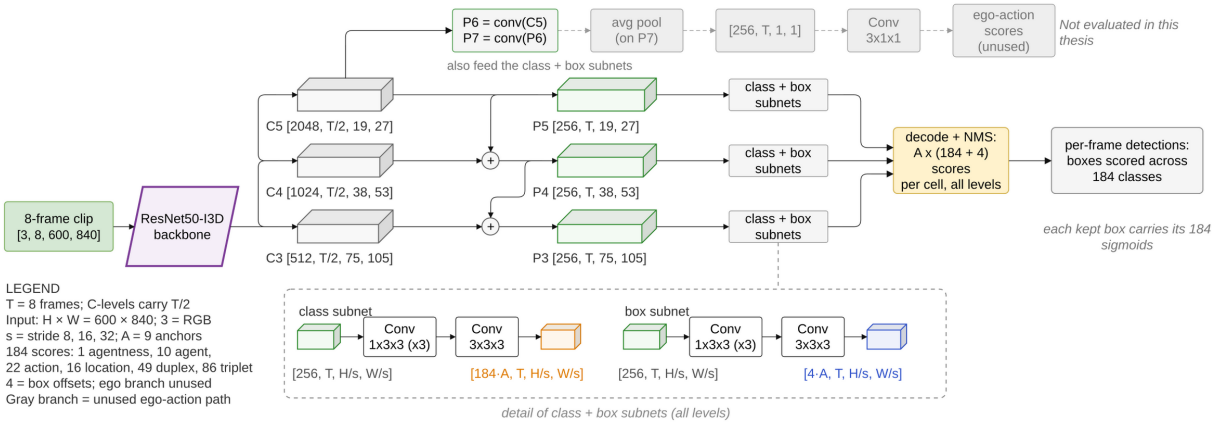


Figure 5: Replicated 3D-RetinaNet with a ResNet50-I3D backbone and feature pyramid [1]. The faded ego-action branch is unused in this thesis.

3.4 Detector Replacement and Score Transfer

YOLOv8x [44], fine-tuned on ROAD-Waymo agent classes with a 1280-pixel input setting, supplies candidate boxes, confidence, and hard agent identity. The project training configuration was informed by the ECCV 2024 Track-1 report [46], with local implementation choices; the selected checkpoint is the recorded epoch-1 model. Candidate generation uses confidence 0.001, NMS IoU 0.7, and at most 300 detections per frame.

Figure 6 details the YOLOv8x detector used for agent localization. The 1280 × 1920 annotation identifies source-frame dimensions; the implementation applies Ultralytics resizing and letterboxing with imgsz = 1280. The reported comparison uses the selected epoch-1 checkpoint from the configured 50-epoch training recipe.

For each YOLO box, Stage 1 finds the baseline detection with highest IoU on the same

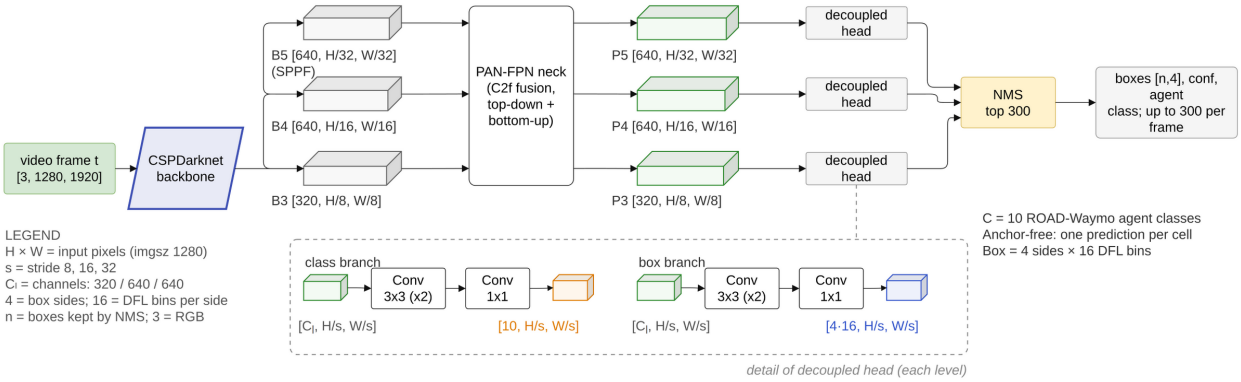


Figure 6: YOLOv8x agent-detector architecture [44].

frame. If IoU is at least 0.5, the action, location, duplex, and triplet scores are copied from that detection. Otherwise, those scores remain zero. Matching is performed independently per YOLO box and does not impose a one-to-one assignment. YOLO supplies the emitted agentness and agent-class detections. No new score-transfer module is trained. Transferred event scores are emitted directly, without multiplication by YOLO confidence.

Writing D_j for the baseline boxes and r_{jk} for their event scores makes the transfer rule explicit:

$$j^* = \arg\max_j \mathrm{IoU}(B_i, D_j), \qquad s_{ik} = \begin{cases} r_{j^*k}, & \mathrm{IoU}(B_i, D_{j^*}) \geq 0.5, \\ 0, & \text{otherwise.} \end{cases} \tag{3.3}$$

The zero case also applies when there are no baseline detections on that frame; k indexes action, location, duplex, or triplet labels.

Across the 36,717 shared frames, approximately 40.6% of 6,163,968 YOLO candidates find a qualifying baseline match. This is the proportion of candidates that receive transferred event scores; it is not ground-truth recall or classification accuracy. Figure 7 illustrates why: an unmatched YOLO box has no baseline event-score source, so its transferred action, location, duplex, and triplet values remain zero. A correctly localized object can consequently lack useful event

predictions. The next stage addresses this dependence by extracting a feature at every candidate box.

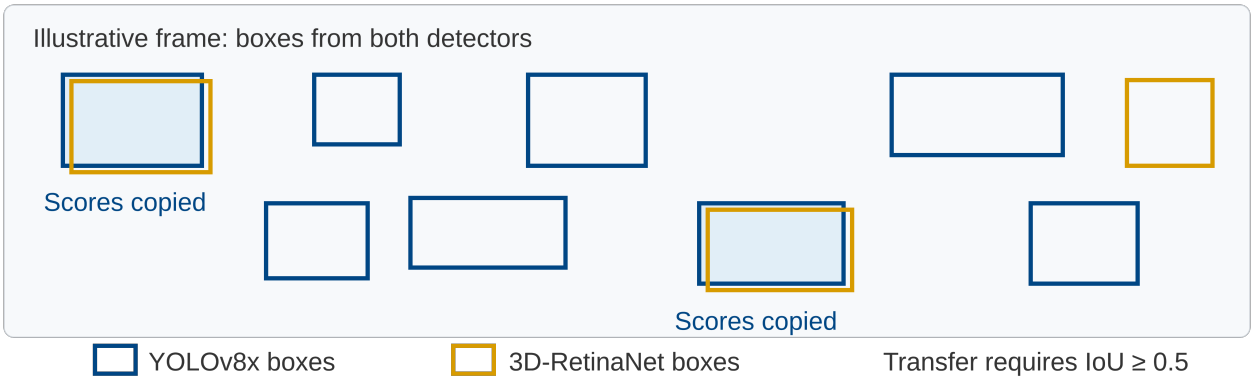


Figure 7: The event-score coverage limitation of overlap transfer. The boxes illustrate matched and unmatched candidates; their counts do not represent the measured match rate.

Figure 8 summarizes the score-transfer path: overlap matching determines whether a YOLO candidate receives the baseline's event scores.

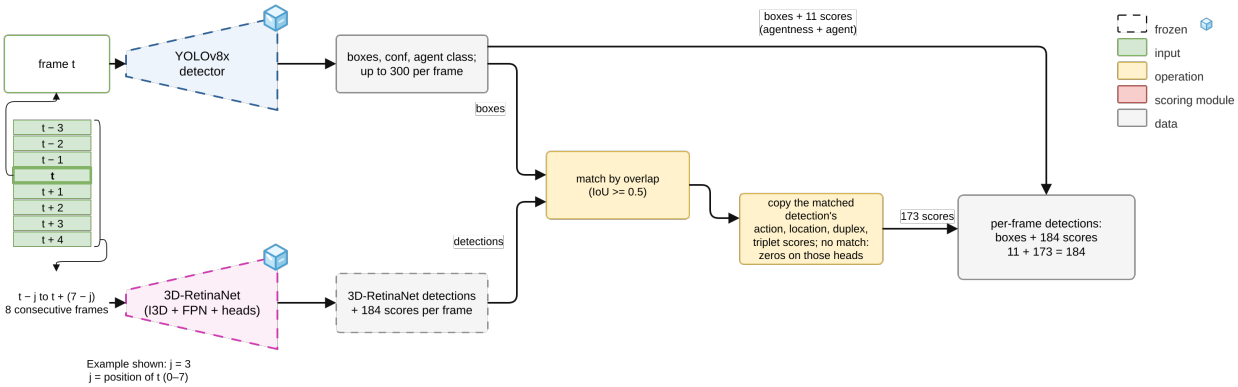


Figure 8: Stage 1 detector replacement and overlap-based score transfer.

3.5 Direct Region Features and Event Classification

Stage 2 replaces overlap-based score transfer with feature extraction at each YOLO box. Figure 9 shows the detector–classifier path: the detector and video backbone remain frozen while

the event head is trained.

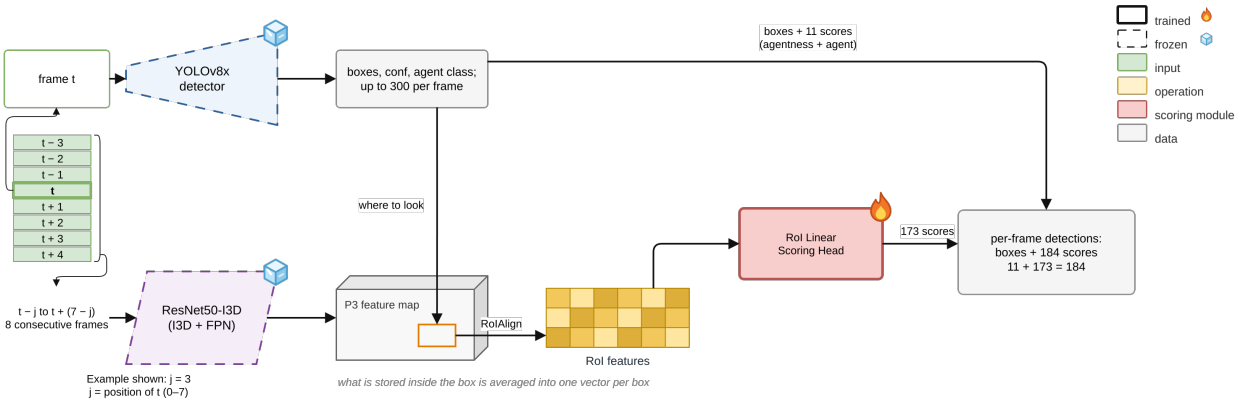


Figure 9: Stage 2 direct region-feature classification.

Stage 2 reads the frozen I3D/FPN P3 feature map at every YOLO box using RoIAlign [13]. Boxes are mapped into feature-map coordinates and pooled to one 256-dimensional vector at the corresponding frame. Temporal interpolation aligns the feature map with the clip length when needed. Unlike score transfer, this procedure can obtain a feature at a location that the baseline detection head did not select.

Figure 10 expands this operation. The implementation rescales box coordinates to the P3 grid, selects the feature slice for the target frame, and applies aligned RoIAlign with a single spatial output bin. Bilinear sampling and averaging produce one 256-dimensional vector for each box. The illustrated sample locations explain the pooling operation; they are not additional detections. Because the map was produced from a video clip, the regional feature can incorporate temporal context even though the readout is associated with one target frame [13].

A linear layer maps each vector to 184 logits. Training uses ground-truth boxes with multi-hot targets, random background boxes, and an additional set of YOLO background candidates. Background targets are zero across the output vector. The reported configuration uses focal classification loss with class-dependent alpha weights and focusing parameter $\gamma = 2$ [12]. The optional

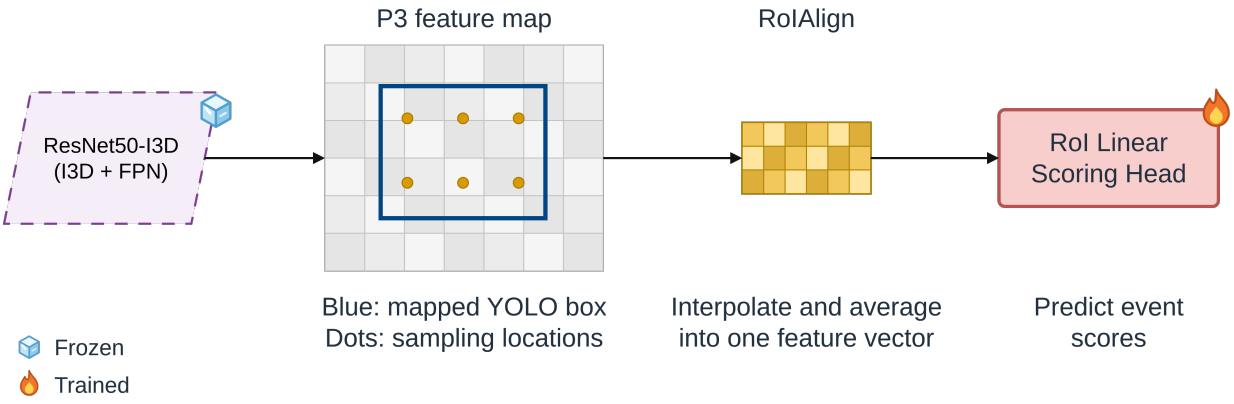


Figure 10: RoIAlign extraction at a proposed box [13], followed by the trained RoI Linear Scoring Head.

logical-penalty coefficient is zero in this configuration.

For the frozen target-frame feature map F_t and proposed box B_i, the learned readout is

$$x_i = \mathrm{RoIAlign}(F_t, B_i) \in \mathbb{R}^{256}, \qquad \ell_i = W_{\mathrm{lin}} x_i + b_{\mathrm{lin}}, \qquad p_i = \sigma(\ell_i). \tag{3.4}$$

Here σ is the elementwise sigmoid and $\ell_i \in \mathbb{R}^{184}$. For a batch of N examples and K trained output labels, the class-weighted focal objective is

$$\begin{aligned} p^{\star}_{ik} &= y_{ik} p_{ik} + (1 - y_{ik})(1 - p_{ik}), \\ \alpha^{\star}_{ik} &= y_{ik} \alpha_k + (1 - y_{ik})(1 - \alpha_k), \\ \mathscr{L}_{\mathrm{focal}} &= -\frac{1}{NK} \sum_{i=1}^{N} \sum_{k=1}^{K} \alpha^{\star}_{ik} (1 - p^{\star}_{ik})^2 \log p^{\star}_{ik}. \end{aligned} \tag{3.5}$$

The fixed α_k values provide class weighting. The 184-output classifier uses $K = 184$; composition-only heads use their 135 composition targets.

During evaluation, the head's action, location, and composition sigmoids are multiplied by

YOLO confidence. For a candidate with confidence q and raw class score p, the emitted score is qp. This gate affects detection ranking; it should not be interpreted as proof of calibrated probability. YOLO remains responsible for the emitted agentness and hard agent class. The Stage 1-to-2 transition changes feature access, the trained classifier, and confidence gating; it is not a feature-access-only ablation.

For these learned heads and the later crop/contextual models, score assembly is

$$s_{ik} = q_i p_{ik}, \qquad k \in \{\text{action, location, duplex, triplet labels}\}. \tag{3.6}$$

The detector confidence q_i is applied once, after any composition MLP or expert blend. Equation 3.3 is the earlier transfer exception.

3.6 Stacked Composition Learning

Stage 3 keeps the Stage 2 feature extractor and full-data head fixed at evaluation. It concatenates 49 ungated primitive scores with the 256-dimensional pooled feature, producing a 305-dimensional input. A two-layer MLP, Linear(305,512) → ReLU → Linear(512,135), predicts 49 duplex and 86 triplet logits. Its sigmoid outputs replace only the composition columns.

The MLP is trained using video-grouped, two-fold out-of-fold predictions. These internal training folds do not replace the held-out validation split; the procedure is detailed below.

At evaluation, the full-training-data head supplies primitive predictions. The MLP receives raw, ungated sigmoids and the pooled feature. Its composition scores are gated by YOLO confidence only at emission. This ordering matches the representation used to train the MLP. Action and location pass through from Stage 2, and YOLO boxes and agent outputs remain unchanged.

Figure 11 details the training-only procedure, following the principle of stacked generalization [47]. Training videos are assigned to two folds. A temporary linear head trained on fold

A supplies predictions for fold B, and the head trained on fold B supplies predictions for fold A. The composition MLP therefore learns from predictions on videos excluded from the corresponding temporary head's training set. Grouping by video prevents adjacent frames from being split between the two folds. The held-out designation applies to these temporary heads; the visual backbone is a shared frozen representation. The temporary heads are discarded after generating the MLP training inputs.

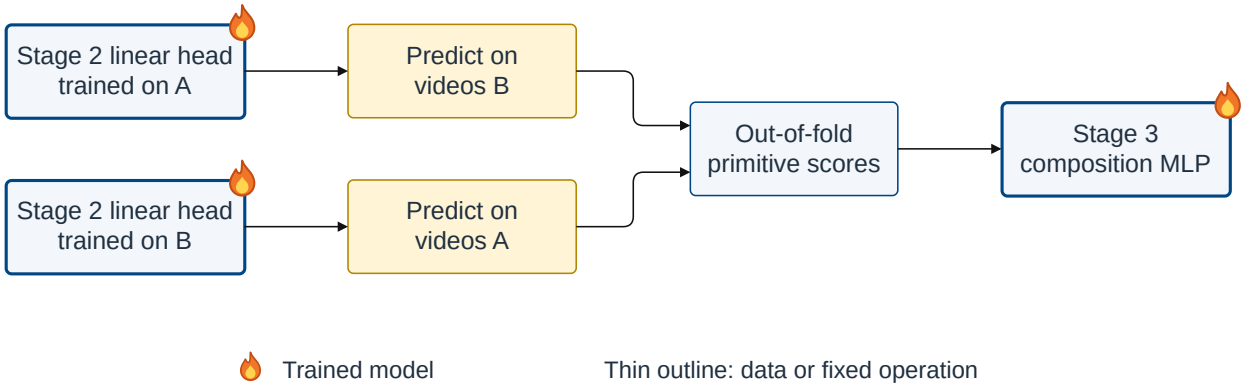


Figure 11: Video-grouped out-of-fold inputs for composition learning, adapted from the stacking principle [47]. A and B are folds of the training split.

Figure 12 follows one candidate from primitive and visual evidence to learned composition scores. The MLP receives 49 ungated scores—agentness plus agent, action, and location—together with 256 regional features. Its 135 outputs replace only the 49 duplex and 86 triplet entries. The feature input supplies visual evidence when primitive scores alone are ambiguous; retaining the remaining outputs makes the Stage 2-to-3 comparison specific to composition learning.

Let $a_i \in [0,1]^{49}$ collect the ungated primitive scores and let $[\cdot,\cdot]$ denote concatenation. The diagram corresponds to

$$h_i = \mathrm{ReLU}(W_1[a_i, x_i] + b_1), \qquad p_i^{\mathrm{comp}} = \sigma(W_2 h_i + b_2). \tag{3.7}$$

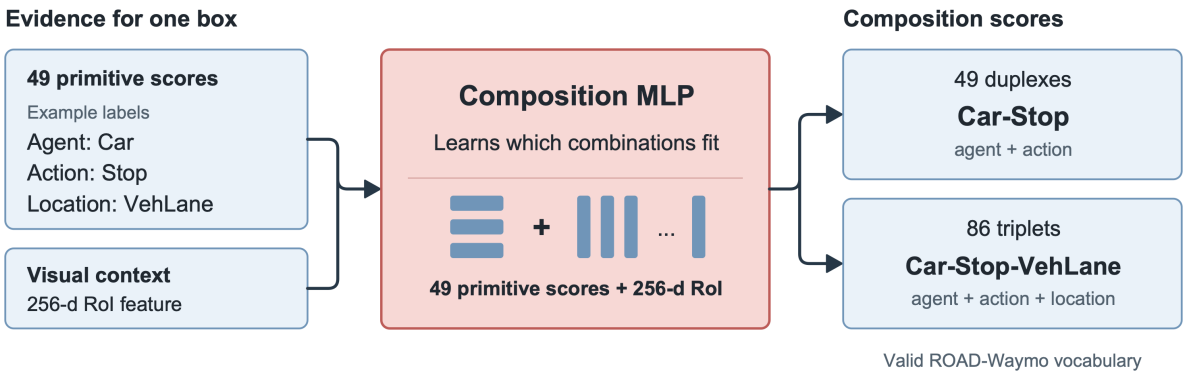


Figure 12: Composition MLP: primitive and visual evidence for one box produces duplex and triplet scores. Car-Stop and Car-Stop-VehLane are illustrative labels, not measured predictions.

For Stage 3, $W_1 \in \mathbb{R}^{512\times305}$ and $W_2 \in \mathbb{R}^{135\times512}$. The 305 inputs are 49 + 256; the 135 outputs are 49 + 86. These are learned scores over the valid composition vocabulary, rather than products of the primitive probabilities.

The output vocabulary includes only the valid benchmark compositions. A nonexistent tuple has no dedicated output column. This vocabulary restriction does not guarantee agreement between independently scored primitives and compositions; cross-head consistency is a separate property.

Figure 13 summarizes the completed Stage 3 architecture. Its composition module builds on the Stage 2 representation while preserving boxes and primitive outputs, providing the reference point for the results in Chapter 4.

3.7 Reproducibility and Label Integrity

The source implementation fixes the 184-column ordering across frozen feature outputs, prediction heads, and evaluation. YOLO boxes are normalized coordinates; baseline boxes stored in an 840 × 600 convention are converted before overlap calculations. These conventions are

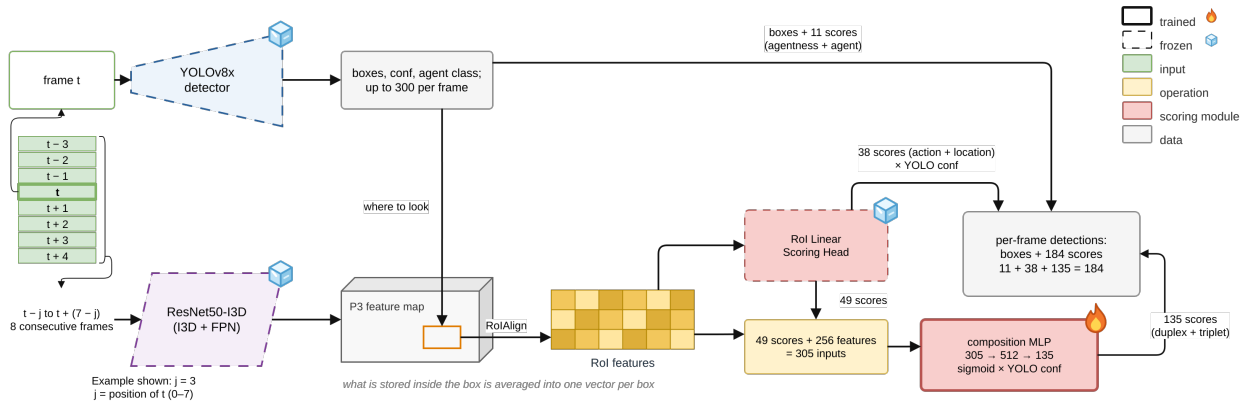


Figure 13: Completed Stage 3 learned-composition architecture.

necessary for meaningful box matching and score assembly.

Valid compositions are derived from benchmark label names and the verified constraints file. The legacy duplex_childs and triplet_childs arrays are not used as authoritative mappings. Experiment records retain the exact class order, candidate and feature identities, seeds, and evaluation manifests.

3.8 Contextual Crop Representation

For each candidate box on target frame t, the extraction path uses eight real frames, $t - 3$ through $t + 4$, with indices clamped at video boundaries. The same spatial region is sampled across the clip; it is not a tracked tube. Default context padding doubles box width and height around the center, subject to image bounds and a minimum crop size. Crops are resized to 224 × 224 before encoding.

Figure 14 shows how one detection defines the clip supplied to the encoder. The same padded rectangle is reused across all eight frames, preserving a consistent spatial window while the scene changes within it. Doubling both width and height yields four times the unpadded area before boundary clipping. Nearby road structure and other agents may therefore enter the representation; this added context is part of the crop representation and must be distinguished from the entire-frame

context introduced below.

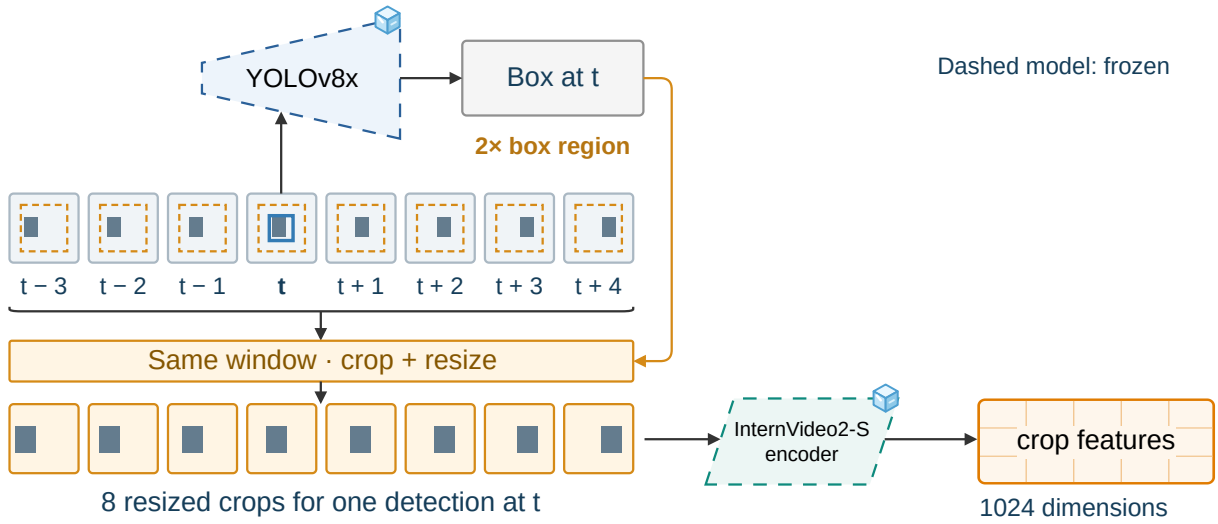


Figure 14: Contextual crop construction for one detection. A fixed padded window is sampled from frames $t - 3$ through $t + 4$, resized, and encoded; this is not a tracked object tube.

The selected model is OpenGVLab/InternVideo2_CLIP_S [41]. A frozen visual tower processes each crop clip. The implementation reads last-block tokens for the target frame and averages its spatial tokens to obtain one 1024-dimensional descriptor. The keyframe is the frame associated with the box and label being evaluated. Away from a boundary, it is the fourth input frame. Because the clip includes frames after t, this is contextual frame detection rather than causal anticipation.

The descriptor is an internal visual representation, not simply the model's final global video–text embedding. A learned projection is therefore part of the classifier. The paired text tower supplies the semantic class directions, while the projection learns how the chosen visual representation connects to them.

Figure 15 distinguishes the pretrained video–text alignment path from the keyframe-token feature tap used in the crop study [40], [41]. The schematic shows patch tokens; the separate class token is omitted. The 1024-dimensional crop descriptor is taken before the released model's pooled

512-dimensional alignment output.

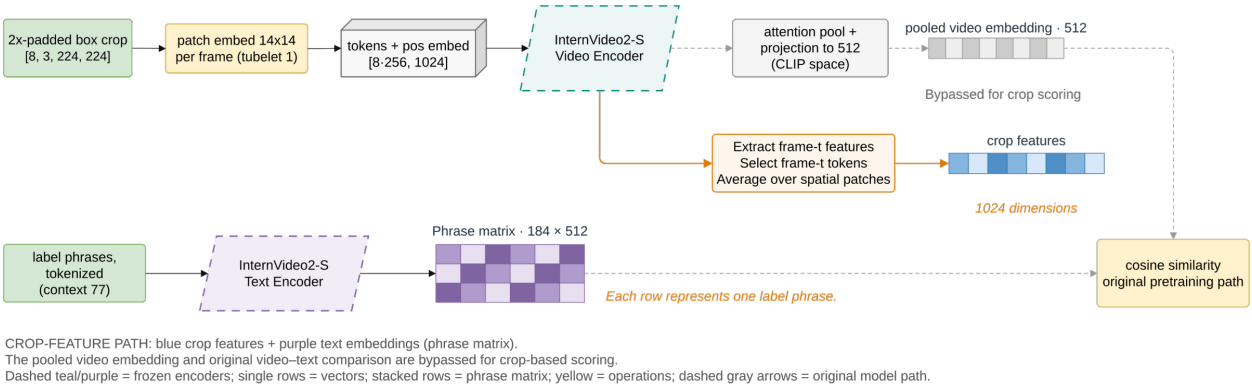


Figure 15: Frozen InternVideo2_CLIP_S towers and the crop-feature tap [40], [41]. The original pooled alignment path is bypassed; the full 184-phrase matrix supplies the semantic bank.

3.9 Phrase-Based Classification

The semantic reference uses explicit event descriptions rather than arbitrary class identifiers. Table 3 gives three entries from the implementation's phrase dictionary. The composition phrase retains the meanings of its parts, while the full set of 184 descriptions remains in benchmark class order. The frozen text encoder processes the dictionary once; the supervised experiment learns how crop features map to those fixed directions [32], [40].

Table 3: Examples of fixed class descriptions.

Benchmark label	Description
Car	a car
Car-Stop	a car stopped
Car-Stop-VehLane	a car stopped in the ego vehicle’s lane

A fixed phrase is assigned to each of the 184 benchmark labels, including separate descriptions for duplexes and triplets. The text encoder embeds these phrases once. Let P be the matrix of normalized phrase vectors, with one row per class. For a crop descriptor x, the classifier computes

$$z = \text{normalize}(Wx + a), \qquad \ell = \exp(\tau)Pz + b. \tag{3.8}$$

Here W and a define the learned visual projection, τ is a learned scalar that controls the positive logit scale, and b contains per-class biases. For the selected implementation, x has 1024 dimensions and the phrase space has 512 dimensions. The phrase matrix and both encoders remain frozen during head training. Sigmoid outputs are trained using the same focal classification-loss formulation as the completed head experiments. The repeated crop-head study uses Adam with learning rate 0.001, ten epochs, and batch size 16,384. Training targets come only from training-split ground-truth boxes and background examples. The study trains seeds 0, 1, and 2 using the final-epoch checkpoints. Separate video-grouped out-of-fold heads supply inputs to the composition models.

The contrastive image–text learning occurs during the encoder's pretraining [32], [40]. Stage 4 uses the resulting fixed phrase embeddings to guide a supervised classifier trained on ROAD-Waymo labels. This experiment tests whether this semantic structure improves recognition of rare compositions.

YOLO continues to supply boxes, confidence, and emitted agent identity. Action, location, and composition scores follow the same emission and confidence-gating convention as the completed hybrid. This keeps the semantic study tied to a fixed detection platform.

Figure 16 summarizes the Stage 4 path, from frozen crop features and class phrases to supervised road-event scores.

3.10 Semantic Crop Composition Models

Stage 5, the Semantic Crop Classifier with Composition MLP, uses an unconstrained linear head on the same frozen crop descriptors. The composition MLP receives 49 raw primitive sig-

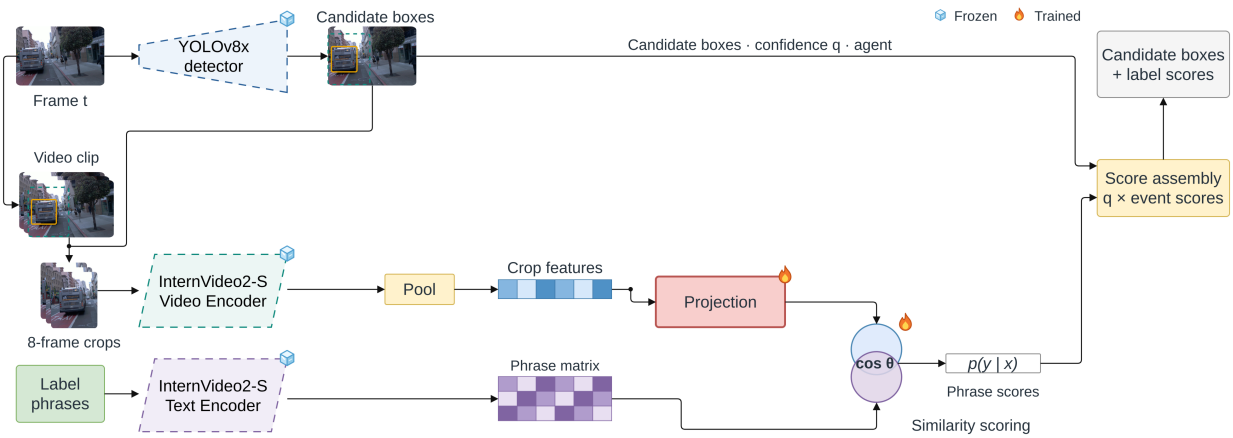


Figure 16: Stage 4 semantic crop classifier.

moids and the 1024-dimensional crop feature, giving 1073 inputs. A 512-unit ReLU hidden layer maps these inputs to 135 duplex and triplet logits. This architecture contains no explicit phrase-score input. Its membership in the semantic-crop family identifies the pretrained representation, not a phrase classifier in its composition path.

Stage 6 adds the 135 phrase-composition sigmoids to the Stage 5 inputs, increasing the input dimension to 1208. It retains the same 512-unit hidden layer and 135-output composition head. The flat head supplies action and location scores in both stages. The distinction between Stage 5 and Stage 6 is therefore the additional phrase-composition evidence available to the MLP.

Writing $x_i^{\text{crop}} \in \mathbb{R}^{1024}$ for the crop feature and $r_i^{\text{phrase}} \in [0,1]^{135}$ for phrase-composition sigmoids, their inputs are

$$u_i^{(5)} = [a_i, x_i^{\text{crop}}] \in \mathbb{R}^{1073},$$
$$u_i^{(6)} = [a_i, x_i^{\text{crop}}, r_i^{\text{phrase}}] \in \mathbb{R}^{1208}. \tag{3.9}$$

Each stage applies its separately trained version of Equation 3.7, with a 512-unit hidden layer and 135 outputs.

Both use video-grouped two-fold out-of-fold head predictions during composition training, and full-training-data heads during evaluation. Composition inputs remain ungated; YOLO confidence is applied once when emitting event detections.

The crop-head study includes three seeds, fixed cached features, identical batch-order generators across classifier variants, and the same ten-epoch Adam configuration described above.

Figure 17 shows the crop-plus-primitive path; Figure 18 adds the phrase branch.

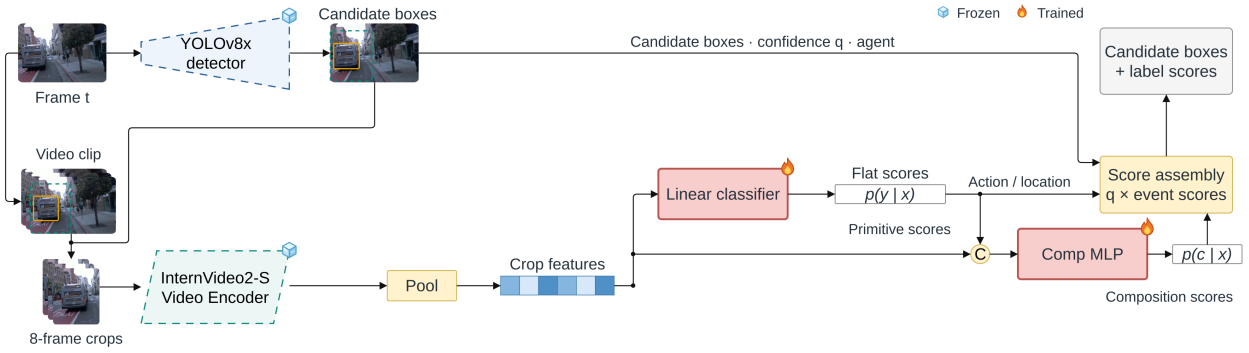


Figure 17: Stage 5 semantic crop classification with Composition MLP. The linear head supplies primitive evidence and action/location outputs; crop features also enter the MLP. The illustrated ROAD-Waymo actor window is not a measured detection [2].

A shuffled-phrase control permutes the phrase-to-label assignment while retaining the same phrase vectors. This is a control on their assignment, not a new encoder or a zero-shot experiment.

3.11 Context Fusion and Scene Attention

The contextual model combines a crop descriptor with actor-region features, entire-frame scene tokens, and box geometry. Its readout directly predicts 184 logits. It is a separate architecture from the Stage 5/6 composition MLP, rather than an additional composition layer appended to those models.

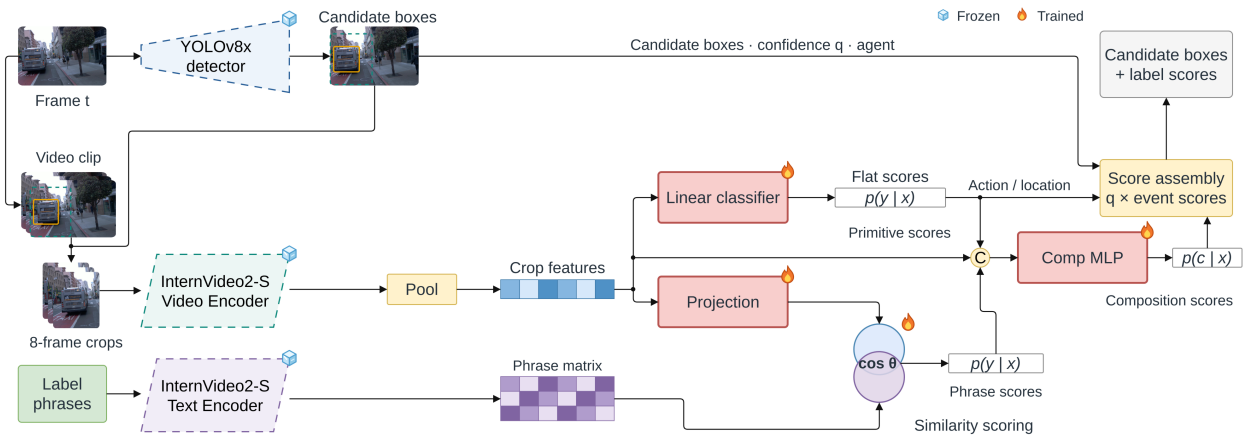


Figure 18: Stage 6 semantic crop classification with phrase fusion and composition. Stage 5 omits the phrase-composition input. Frozen encoders provide cached features. The ROAD-Waymo imagery illustrates candidate-centered processing [2].

A full eight-frame clip passes through the same frozen visual encoder used for crops. For the original encoder, the keyframe supplies a 16×16 spatial grid of 1024-dimensional tokens. A 7×7 RoIAlign operation followed by spatial averaging gives one actor-region descriptor. Pooling the full grid to 4×4 supplies 16 scene tokens. These tokens preserve coarse spatial context outside the candidate crop. The candidate's normalized coordinates and derived center, width, and height form the geometry vector $(x_1, y_1, x_2, y_2, c_x, c_y, w, h)$.

A residual crop adapter and projection produce $c \in \mathbb{R}^{512}$. Independent projections map the actor-region descriptor and scene tokens to the same width, giving r and S. A learned MLP maps geometry to $g \in \mathbb{R}^{512}$. The mean-pool variant averages the projected scene tokens. The scene-attention variant instead uses the actor-conditioned query $c + g$, with scene tokens as keys and values [48]:

$$s = \mathrm{MHA}(c + g, S, S). \tag{3.10}$$

Eight attention heads select scene evidence relevant to the candidate. A fusion MLP maps the

concatenation $[c, r, s, g]$, of dimension 2048, to 512 dimensions. A residual crop connection and layer normalization produce the pre-language visual representation

$$v = \mathrm{LN}\left(c + \mathrm{MLP}([c, r, s, g])\right). \tag{3.11}$$

Geometry enters the downstream head; it is not inserted into the frozen encoder as pixels or additional pretrained tokens.

For comparison, mean pooling uses $s = \frac{1}{16}\sum_{m=1}^{16} S_m$. In the attention variant, each of the eight heads computes

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d_h}}\right)V, \qquad d_h = 512/8 = 64. \tag{3.12}$$

For scene attention, $Q = (c + g)W_Q$, $K = SW_K$, and $V = SW_V$ within each head. Concatenating the head outputs and applying the output projection gives s. The softmax runs over the scene-token positions. Figures 19 and 20 show the two scene-summary choices with the same crop residual and language-fusion path.

3.12 Language Fusion and Contrastive Supervision

Let P denote the frozen ordered phrase bank. A trainable residual text adapter produces $T = P + \mathrm{MLP}(\mathrm{LN}(P))$ for the original 512-dimensional bank. Language cross-attention uses v as query and T as keys and values. Its output is scaled by a learned sigmoid gate, added to v, and normalized to obtain the joint representation j. Separately, normalized v and each row of T produce 184 cosine-similarity scores, denoted u. The final classifier is

$$\ell = \mathrm{MLP}_{1208\to512\to184}([j, c, u]). \tag{3.13}$$

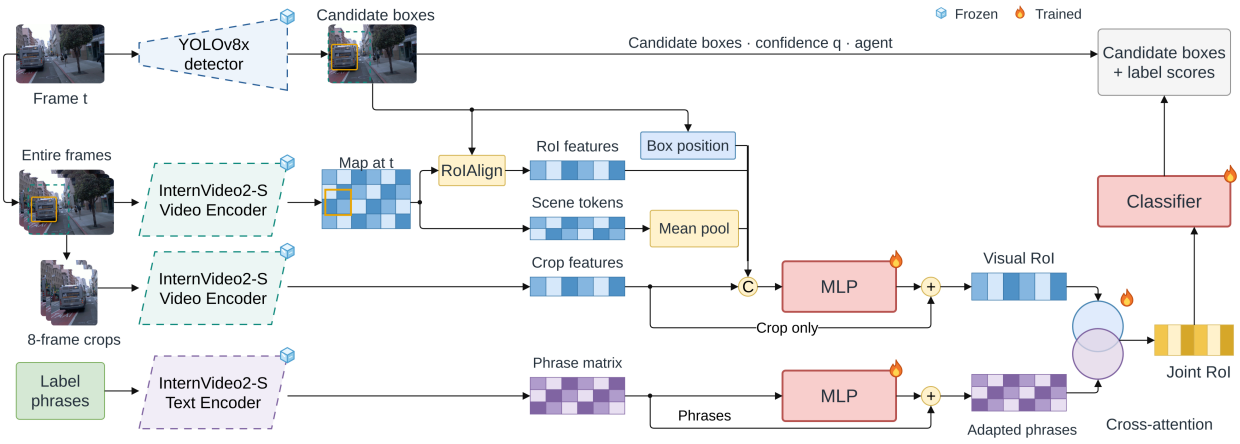


Figure 19: Context fusion with mean-pooled scene tokens and language cross-attention. Crop, actor-region, scene, and box-position evidence feed the visual fusion MLP. The crop-only skip preserves the projected crop representation.

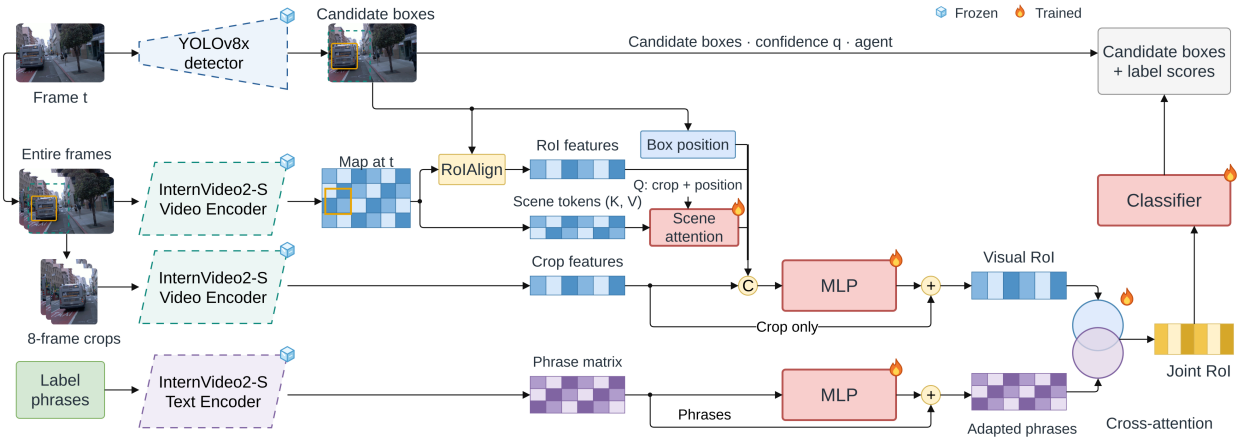


Figure 20: Context fusion with scene attention. The query is crop plus position; projected scene tokens supply keys and values. Language cross-attention is a separate operation after visual fusion.

The direct crop input remains present alongside joint and phrase evidence. Both the classification-only and contrastive variants contain language attention and phrase scores. Classification-only means that the auxiliary contrastive coefficient is zero, not that language is absent.

The joint feature and phrase-similarity branch can be written as

$$j = \mathrm{LN}(v + \sigma(\beta)\,\mathrm{MHA}(v, T, T)),$$
$$u_k = \frac{v^{\mathrm{T}} T_k}{\|v\|_2 \|T_k\|_2}, \qquad k = 1, \ldots, 184. \tag{3.14}$$

The learned scalar β gates the language residual. Equation 3.12 also describes each language-attention head, now with projected visual RoI queries and adapted phrase keys and values. The diagrams summarize this readout as a classifier; its explicit inputs remain $[j, c, u]$.

The corrected contrastive objective aligns the pre-language visual representation with adapted phrases across all 184 labels. For row i, let Y_i be its positive-label set and let I_+ contain rows with at least one positive label. With temperature $\theta = 0.07$, the auxiliary loss is

$$\mathscr{L}_{\mathrm{con}} = -\frac{1}{|I_+|} \sum_{i \in I_+} \frac{1}{|Y_i|} \sum_{k \in Y_i} \log \frac{\exp(\cos(v_i, T_k)/\theta)}{\sum_{m=1}^{184} \exp(\cos(v_i, T_m)/\theta)}. \tag{3.15}$$

All-zero background rows are excluded from this auxiliary, but remain in classification training. The full objective is $\mathscr{L}_{\mathrm{cls}} + 0.001\,\mathscr{L}_{\mathrm{con}}$. The auxiliary updates both visual and text adapters; frozen encoder outputs and the original phrase bank remain fixed. It acts before language attention and does not compare classifier logits with phrases. At inference the complete phrase bank is available without ground-truth labels.

An earlier auxiliary used only 86 triplet prototypes from the original frozen bank. The corrected recipe changes both label scope and gradient flow into the text adapter. Comparisons

between these recipes therefore do not isolate label scope alone. The main contrastive contextual results in this thesis use the corrected all-184 recipe.

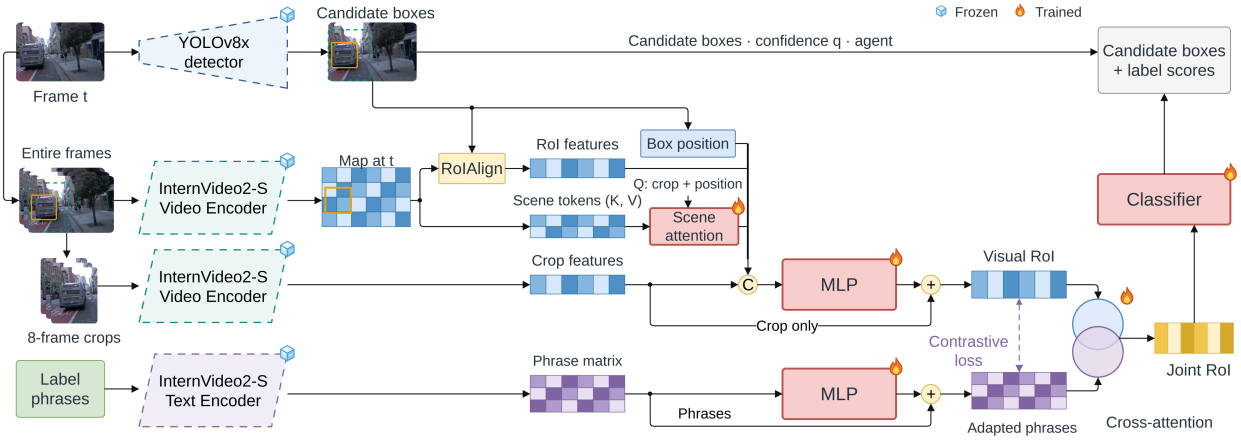


Figure 21: Contextual scene attention with language fusion and all-184 contrastive supervision. The contrastive branch aligns the pre-language visual representation with adapted phrases during training. ROAD-Waymo imagery is illustrative [2].

3.13 Dynamic Class Balancing

Dynamic class balancing (DCB), motivated by FRCB [22], replaces the statically weighted focal classification loss with a difficulty-weighted binary cross-entropy. For predicted probability $p_{ik} = \text{sigmoid}(\ell_{ik})$ and binary target y_{ik}, the local implementation uses

$$B_{ik} = 0.5 + y_{ik}(1 - p_{ik}) + h_k, \qquad \mathcal{L}_{\text{DCB}} = \frac{1}{184N} \sum_{i,k} B_{ik}\, \text{BCE}(\ell_{ik}, y_{ik}). \tag{3.16}$$

Here h_k is the mean positive-label residual for class k, accumulated over prior batches in the run. It is zero until a positive example has been observed. The weights are detached from the computation graph, and the running residual sums and positive counts update after forming the current loss. History persists across epochs and is saved in checkpoints. Negative entries also receive the historical class term.

For batch index b, the historical term excludes the current batch:

$$h_k^{(b)} = \begin{cases} \dfrac{\sum_{a<b} \sum_{i \in \mathcal{B}_a} y_{ik}(1 - p_{ik})}{\sum_{a<b} \sum_{i \in \mathcal{B}_a} y_{ik}}, & \sum_{a<b} \sum_{i \in \mathcal{B}_a} y_{ik} > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{3.17}$$

Predictions in this history are those observed when each earlier batch was processed. This is a running positive-label residual, not an error rate recomputed over the current validation set.

DCB replaces both the focal modulator and its class-alpha weighting; these factors are not multiplied into the DCB objective. The comparison is therefore dynamic DCB versus an already class-weighted focal baseline. The implementation averages over box–class entries, whereas the published equation sums over classes and averages over boxes. The history, zero-positive guard, and reduction are explicit local choices. This is an FRCB-inspired loss experiment within the contextual head, not reproduction of FRCB's full architecture. DCB changes training only and adds no label-dependent inference operation.

3.14 AP-Selected Expert Blending

The Contextual Semantic Expert Blend (CSEB) combines two separately trained scene-attention heads. The classification expert uses classification loss alone. The contrastive expert uses the same classification loss plus the all-184 auxiliary. Both include language fusion and predict all label groups. After training, both experts are frozen. Their composition probabilities are combined as

$$p_{\text{blend},k} = (1 - w)p_{\text{cls},k} + wp_{\text{con},k}, \qquad k \in \{\text{duplex and triplet labels}\}. \tag{3.18}$$

A single scalar per seed is selected from $\{0, 0.125, \ldots, 1\}$ by maximizing triplet crop AP on 45 reserved training videos. Ties select the smaller contrastive weight. The weight stays fixed during

inference. This is probability mixing, not a learned sample-dependent gate or logit averaging. The 49 primitive and agentness scores come from the classification expert before final assembly. Emitted agentness and agent identity still come from YOLO, and event probabilities are multiplied by YOLO confidence once.

Equivalently, the selection rule is

$$w^* = \min\left(\arg\max_{w\in\{0,0.125,\ldots,1\}} \mathrm{mAP}^{\mathrm{crop}}_{\mathrm{triplet},\,45\ \mathrm{videos}}(w)\right). \tag{3.19}$$

This internal crop metric selects the blend and is distinct from the detector metric in Equation 3.2. Figure 22 illustrates the arithmetic for one synthetic composition score; Figures 23 and 24 show the focal and DCB expert pairs.

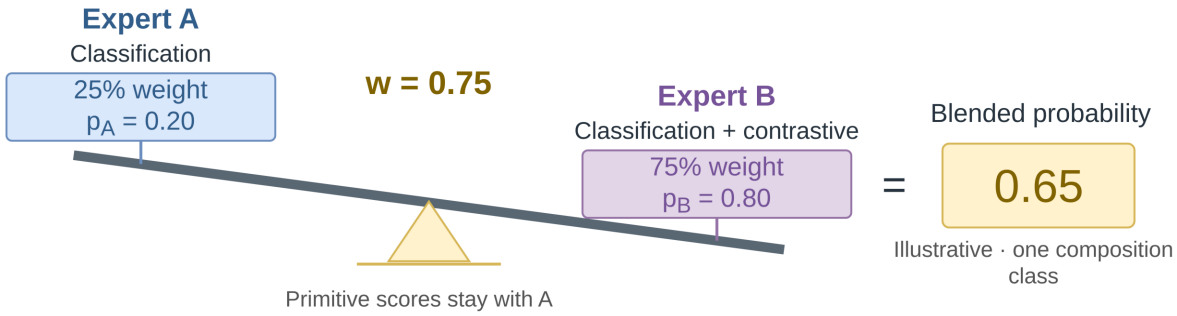


Figure 22: Illustrative expert probability blend from the defense slideshow. Expert A is the classification expert and Expert B adds contrastive supervision. With weight 0.75, the synthetic score is 0.25(0.20)+0.75(0.80)=0.65. Primitive scores remain with A.

3.15 Encoder Scaling and Training Partitions

The final CSEB uses the released InternVideo2-CLIP-1B system [49]. The visual checkpoint has 40 blocks of width 1408, compared with 24 blocks of width 1024 in the original InternVideo2_CLIP_S release. The original release name does not imply a conventional small visual

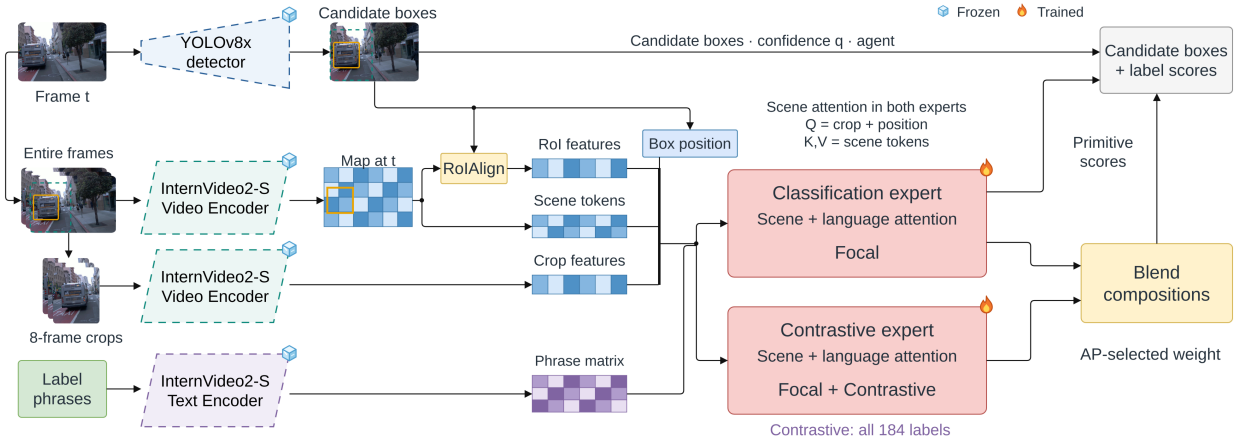


Figure 23: Contextual expert blend with focal training. Both experts use scene and language attention; the contrastive expert additionally uses all-184 alignment. Only composition probabilities are blended.

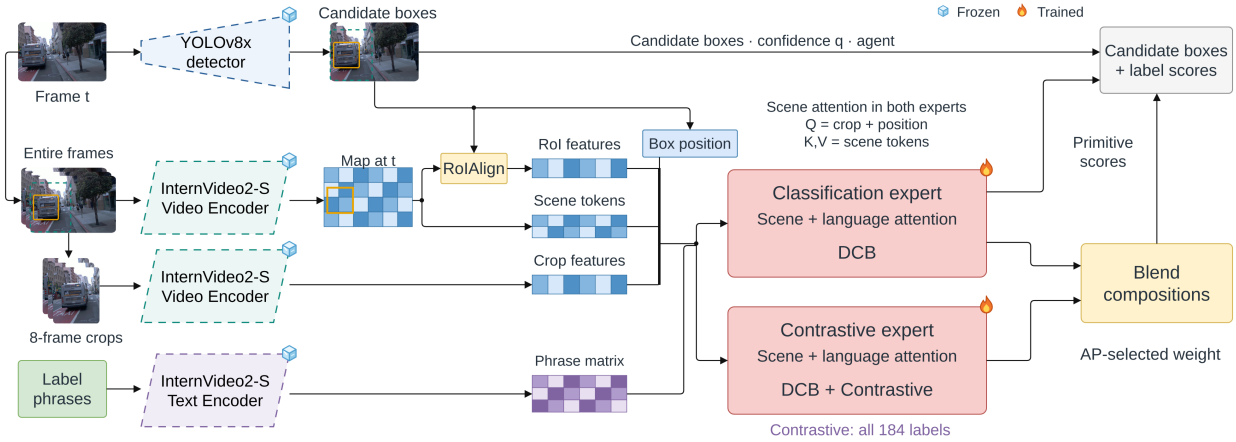


Figure 24: Contextual expert blend with DCB training and the original encoders. DCB replaces focal classification in both experts. The contrastive auxiliary and inference topology are retained.

transformer. The scaled system also changes the pretrained text checkpoint to the publisher's paired InternVL/Llama-based text implementation [50], producing 768-dimensional phrase embeddings. A trainable 768 → 512 projection connects these embeddings to the contextual head. The downstream internal width remains 512. The system-level text-encoder name does not imply that the text model itself has one billion parameters.

The scaled caches use BF16 encoder computation, float32 preprocessing and pooling, and float16 feature storage. The original cache path used FP16 preprocessing and encoder computation. These changes, together with the paired visual/text checkpoints and text projection, make scaling a system comparison rather than an isolated parameter-count ablation. The visual and text encoders remain frozen during contextual-head training.

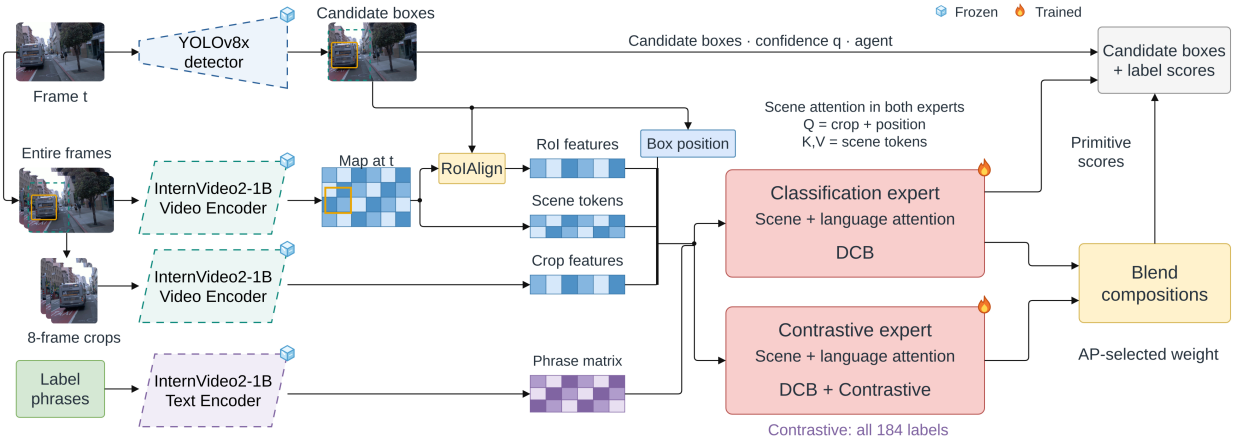


Figure 25: Scaled Contextual Semantic Expert Blend. The classification expert uses DCB; the contrastive expert adds all-184 alignment. Both use scene and language attention. Only composition probabilities are blended. ROAD-Waymo imagery is illustrative [2].

The contextual studies partition the 600 training videos into 420 expert-training videos, 90 development videos, and 90 reserved gate videos. The global blend uses the same 45-video selection half of the gate partition across studies. Expert training uses three epochs, batch size 256,

learning rate 10^{-4}, and weight decay 0.01. Development triplet crop AP selects the checkpoint among the three epochs; final detector validation does not select the checkpoint or blend weight. Seeds 0, 1, and 2 are repeated for each reported contextual condition.

The original crop-head study and contextual studies do not use identical training schedules or partitions. Their comparison evaluates complete systems, while comparisons within a contextual study provide narrower controls. Moreover, the shared ROAD-Waymo validation results informed subsequent research choices across the project. This remains a development benchmark, even though checkpoint and blend selection use internal training partitions.

The final detector evaluation uses the locked 36,717-frame validation manifest and YOLO candidate identities. Reported development crop AP is a checkpoint-selection measure and is not substituted for detection AP. Results retain all six output groups and the fixed 47-class tail, 28-class deeper tail, and complementary 39-class common group. Means and sample standard deviations across seeds characterize training variability under these fixed data, rather than uncertainty across independently sampled driving environments.

3.16 Worked Example: From One Candidate to an Event Score

Consider the annotated Car-Stop-VehLane instance in Figure 1. Its image and label are real ROAD-Waymo evidence, but the probabilities in this example are deliberately synthetic. They explain the computation and are not predictions measured on that frame. At evaluation, YOLO must first supply a candidate box; the ground-truth annotation does not create a detection.

For a candidate on frame t, the same padded spatial window is sampled across frames $t-3$ through $t+4$. A separate entire-frame pass supplies the spatial map used for actor-region RoIAlign and scene tokens. The crop asks what the local actor looks like over the clip; the region feature locates it within the full scene; the scene tokens provide surrounding road evidence. Box geometry

supplies location explicitly. A single still image cannot establish that the vehicle is stopped, and the contextual window is not a causal forecast.

In Stage 3, the 49 primitive scores and 256 RoI features form a 305-dimensional vector. For this car, the entries associated with Car, Stop, and VehLane are only three coordinates of that vector. The MLP sees all primitive scores and the visual features, learns a 512-dimensional hidden representation, and emits 135 composition scores. It does not simply multiply the three named primitive probabilities. Stages 5 and 6 use the corresponding 1073- and 1208-dimensional inputs defined in Equation 3.9.

In CSEB, each expert instead fuses crop, region, scene, and geometry evidence into a visual RoI. The scene-attention query is the projected crop plus geometry. Language attention uses the resulting visual RoI to query all adapted phrases, including the phrase for Car-Stop-VehLane. Each expert then predicts the full label vector directly. The two experts can disagree about the same candidate despite using the same frozen feature sources because their learned parameters and training objectives differ.

Table 4: Synthetic calculation for one composition label.

Quantity	**Illustrative value**
YOLO confidence q	0.90
Classification-expert composition score p_{cls}	0.20
Contrastive-expert composition score p_{con}	0.80
Fixed contrastive mixing weight w	0.75
Mixed composition $(1 - w)p_{\text{cls}} + wp_{\text{con}}$	0.65
Emitted composition score $q\,p_{\text{blend}}$	0.585

Table 4 extends Figure 22 through detector assembly. The 0.585 score ranks this candidate; it is not a calibrated event probability. AP depends on spatial matching and class-specific ranking across all evaluated frames. A correct label with insufficient box overlap is still a false positive.

CHAPTER 4: Results and Discussion

Table 5 reports the completed Stages 0–3 under the shared-frame protocol. The values are validation f-mAP percentages at IoU 0.5 on the same 36,717 frames. Stage 0 uses its own baseline candidates; Stages 1–3 reuse the fixed YOLO candidates. The results are project measurements.

Table 5: Completed-stage validation f-mAP (%).

Output	Stage 0	Stage 1	Stage 2	Stage 3
Agentness	35.36	65.36	65.36	65.36
Agent	16.67	35.25	35.25	35.25
Action	13.23	12.86	15.92	15.92
Location	12.77	13.36	19.60	19.60
Duplex	11.32	10.04	11.13	15.52
Triplet	7.54	6.81	6.65	9.73

The detector substitution raises agent f-mAP from 16.67% to 35.25%. However, improved agent detection does not automatically improve compositions. Stage 1 reaches 6.81% triplet f-mAP, below the Stage 0 value of 7.54%, while retaining the baseline-detection matching dependency. This comparison does not isolate that dependency's contribution from changes in candidates and score construction.

Stage 2 removes that matching dependency and improves action and location f-mAP to 15.92% and 19.60%. Its free composition columns reach 11.13% duplex and 6.65% triplet f-mAP. This separates two observations: the region features support improved primitive classification, but the independently learned composition scores still leave a substantial gap.

4.1 Effect of Learned Composition

Stage 3 increases duplex f-mAP from 11.13% to 15.52% and triplet f-mAP from 6.65% to 9.73%, while boxes, agentness, agent, action, and location outputs remain unchanged. The changes are 4.39 and 3.08 percentage points for duplex and triplet, respectively. Relative to Stage 0, Stage

3 is higher on all six output groups in this comparison.

The auxiliary composition comparison provides additional context. An MLP using primitive scores alone reaches 13.37% duplex and 8.93% triplet f-mAP; adding the 256-dimensional visual feature raises those values to 15.52% and 9.73%. Fixed minimum composition gives 10.36% and 5.52%, while corrected single-gate product composition gives 10.58% and 6.05%. These measurements support learning composition from both primitive evidence and visual features in this setting. They do not establish a universal advantage over every possible structured model.

4.2 Rare-Composition Performance and Limitations

Table 6 separates the overall triplet result from the training-count-defined tail. The $z < 0$ group contains 47 classes and the $z < -0.5$ group contains 28 of those same classes. These memberships remain fixed across stages.

Table 6: Overall and tail triplet AP (%).

Triplet group	Stage 0	Stage 1	Stage 2	Stage 3
All 86	7.54	6.81	6.65	9.73
$z < 0$ (47)	4.14	3.43	2.60	4.53
$z < -0.5$ (28)	2.88	2.19	1.47	3.31

Stage 3 recovers the tail losses seen in Stages 1 and 2 and exceeds the Stage 0 primary-tail result by 0.39 percentage points.

The controlled Stage 2-to-3 comparison is particularly informative for the rare-event question: primary-tail AP rises from 2.60% to 4.53%, a gain of 1.93 percentage points, while deeper-tail AP rises from 1.47% to 3.31%, a gain of 1.84 points. Candidate boxes and primitive outputs are fixed in this comparison, so these gains accompany the learned-composition replacement rather than a new detector. Nevertheless, primary-tail AP remains 4.53%, compared with 9.73% over all triplets. The deeper-tail value is 3.31%. This is the central empirical motivation for Stage 4: gains

in overall composition recognition leave rare-event performance low.

4.3 Semantic Crop Classification and Composition

Tables 7 and 8 summarize the original-encoder crop models as means across seeds 0, 1, and 2. These rows use the shared detector evaluation and must not be confused with later single-seed ROAD-adapted checkpoints. Agentness and agent outputs remain YOLO-derived. Stage 4 uses phrase classification; Stage 5 uses flat primitive scores and crop features; Stage 6 adds phrase-composition evidence to the composition MLP.

Table 7: Original-encoder crop-model detector f-mAP (%), three-seed means.

Output	Stage 4	Stage 5	Stage 6
Agentness	65.36	65.36	65.36
Agent	35.25	35.25	35.25
Action	18.24	18.82	18.82
Location	19.64	22.50	22.50
Duplex	15.34	17.69	18.12
Triplet	9.07	11.23	11.28

Table 8: Crop-model triplet groups (%), three-seed means.

Group	Stage 4	Stage 5	Stage 6
All 86	9.07	11.23	11.28
Common 39	13.03	17.33	17.43
Tail 47	5.77	6.16	6.17
Deeper tail 28	4.66	4.77	4.57

Table 9: Matched crop-head controls (%), three-seed means.

Classifier	Triplet	Tail 47	Deep 28
Flat linear	8.42	3.54	2.78
Real phrases	9.07	5.77	4.66
Shuffled phrases	4.65	0.71	0.22

The same-feature crop controls in Table 9 distinguish phrase classification from changing the encoder. Real phrases exceed the flat linear head by 0.65 points in triplet mAP and 2.23 points in tail mAP. Permuting the same phrase bank across class labels reduces both substantially. The real-versus-shuffled comparison supports the usefulness of the phrase-to-label assignment in this supervised classifier. The comparison against a flat linear head also changes classifier constraints and normalization, so it does not isolate language independently of head design.

The phrase classifier reaches 9.07% triplet and 5.77% tail mAP. Relative to the Stage 3 I3D composition model, overall triplet mAP is lower by 0.67 percentage points, while tail mAP is higher by 1.25 points. This transition changes both visual representation and head design, so the difference does not isolate language.

Stage 5 raises overall triplet mAP to 11.23% and tail mAP to 6.16%. Stage 6 reaches 11.28% and 6.17%, respectively. The additional phrase-composition inputs therefore change the overall mean by only 0.05 points and the primary-tail mean by less than 0.01 points, while deeper-tail mAP decreases by 0.21 points. These small, mixed changes do not support a large standalone benefit from adding phrase scores to the crop composition MLP.

4.4 Contextual Fusion, Contrastive Learning, and Loss Choice

Table 10 follows the contextual branch using the original encoders. Each row is a three-seed mean. All listed heads already contain language attention and phrase-similarity inputs. The "contrastive" designation identifies the additional all-184 alignment objective. These models use the contextual training partition and direct 184-output readout described in Chapter 3.

Mean-pool contextual fusion reaches 11.74% triplet mAP, but its 5.47% tail and 3.58% deeper-tail means are below the crop Stage 6 means. Context does not improve every group automatically. Replacing mean pooling with actor-guided scene attention increases triplet mAP to

Table 10: Original-encoder contextual triplet mAP (%), three-seed means.

Condition	All 86	Tail 47	Deep 28	Common
Mean pool, focal	11.74	5.47	3.58	19.29
Attention, focal	12.02	6.19	4.48	19.06
Attention, focal + contrastive	12.12	5.79	3.85	19.75
Attention blend, focal	12.58	6.49	4.74	19.93
Attention, DCB	12.90	5.84	4.08	21.40
Attention, DCB + contrastive	12.88	5.81	4.20	21.40
Attention blend, DCB	13.43	6.23	4.54	22.10

12.02%, tail to 6.19%, and deeper tail to 4.48%. This is the appropriate within-family attention comparison, with increases of 0.28, 0.72, and 0.89 points.

Adding the corrected all-184 contrastive objective to the attention head increases triplet mAP by 0.10 points, but decreases tail mAP by 0.40 and deeper-tail mAP by 0.63 points. The global probability blend of the classification and contrastive experts reaches 12.58% triplet, 6.49% tail, and 4.74% deeper-tail mAP. Both experts use language, so this result supports complementarity between their predictions rather than an isolated language effect.

Replacing focal loss with DCB increases the standalone classification expert's triplet mAP from 12.02% to 12.90%, while tail decreases from 6.19% to 5.84%. The DCB blend reaches 13.43% triplet mAP, 0.84 points above the focal blend, but tail and deeper-tail mAP fall by 0.25 and 0.20 points. Thus, the original-encoder DCB study improves overall composition recognition without meeting the corresponding tail-improvement expectation. These losses are retained in the interpretation of the final system.

4.5 Original-Encoder Language Controls

The focal scene-attention study compares real phrases with two controls. The bypass condition disables explicit downstream language paths. The random-bank condition retains active language modules and replaces the phrase bank with seeded random unit vectors. Both controls

retain the visual cache, data partition, head-training schedule, and detector evaluation. The contrastive coefficient is zero in these runs.

Table 11: Original-encoder language controls: mean ± sample SD (%), three seeds.

Condition	Triplet	Tail 47	Deep 28
Real phrases	12.02 ± 0.42	6.19 ± 0.59	4.48 ± 1.15
Language bypass	11.88 ± 0.28	5.76 ± 0.62	4.27 ± 0.89
Random prototypes	11.88 ± 0.42	5.65 ± 0.63	4.02 ± 0.72

Real phrases exceed bypass by 0.43 points and random prototypes by 0.53 points on the tail mean. The corresponding overall gains are 0.14 and 0.15 points. This provides evidence that the explicit phrase branch can help rare classes in this configuration, while overall changes are modest. Bypass changes the active computation, and random prototypes differ from real phrases in both geometry and content. The comparisons therefore do not isolate linguistic meaning alone. With three seeds, a small or statistically unresolved difference must not be interpreted as equivalence. These original-encoder focal controls also do not establish the same effect for the scaled DCB blend.

4.6 Scaled Contextual Semantic Expert Blend

Tables 12 and 13 report the final paired-encoder system. The classification and contrastive experts use the same frozen scaled caches and matched training procedure. The blend combines their composition probabilities while preserving primitive outputs from the classification expert. The reported values are detector f-mAP, not the development crop AP used to select checkpoints and weights.

CSEB reaches 15.75% triplet, 9.00% tail, and 6.35% deeper-tail mAP. Its triplet scores for seeds 0, 1, and 2 are 16.06%, 15.54%, and 15.65%. The blend exceeds both standalone experts on triplet, tail, and deeper-tail mAP within each matched seed. Compared with the classification ex-

Table 12: Scaled contextual detector f-mAP (%), three-seed means.

Output	Classification	Contrastive	CSEB
Agentness	65.36	65.36	65.36
Agent	35.25	35.25	35.25
Action	24.75	24.85	24.75
Location	29.64	29.88	29.64
Duplex	23.37	23.53	24.36
Triplet	15.03	15.08	15.75

Table 13: Scaled contextual triplet groups: mean ± sample SD (%).

Condition	Triplet	Tail 47	Deep 28
Classification	15.03 ± 0.21	8.40 ± 0.58	6.05 ± 0.54
Contrastive	15.08 ± 0.30	8.53 ± 0.63	5.99 ± 0.53
CSEB	15.75 ± 0.27	9.00 ± 0.64	6.35 ± 0.57

pert, its mean gains are 0.72 points overall, 0.60 on the tail, and 0.30 on the deeper tail. In contrast, the standalone contrastive expert differs from classification by only 0.06 points overall, 0.14 on the tail, and −0.06 on the deeper tail. The strongest evidence here concerns expert complementarity, rather than a large isolated contrastive benefit.

Relative to the original-encoder DCB blend, the scaled CSEB improves triplet mAP from 13.43% to 15.75%, tail mAP from 6.23% to 9.00%, and deeper-tail mAP from 4.54% to 6.35%. Common-class mAP also rises from 22.10% to 23.88%. These are system-level changes: both encoder checkpoints, the text input projection, and cache precision differ. They do not identify the causal contribution of visual parameter count alone.

The contribution of each disjoint frequency group to the overall change can be computed because the 86 triplets comprise 47 tail and 39 common classes:

$$\Delta\mathrm{mAP}_{86} = \frac{47}{86}\Delta\mathrm{mAP}_{47} + \frac{39}{86}\Delta\mathrm{mAP}_{39}. \tag{4.1}$$

Using the unrounded means, the 2.32023-point overall improvement contains 1.51155 points from tail classes and 0.80868 points from common classes. Relative to the original-encoder DCB blend, approximately 65% of the triplet mAP improvement comes from tail classes, while common-class performance also improves. This is an arithmetic decomposition across class groups, not a causal decomposition of architectural components. The deeper-tail group is nested within the tail and must not be added again.

4.7 Comparison with Published ROAD-Waymo Results

Table 14 separates published reference results from the local matched-frame study. The 3D-RetinaNet benchmark row is from ROAD-Waymo's validation comparison [2]. The other external rows are the best-result evaluations reported in FRCB Table 3 [22]; they are not new replications of the original papers. The CSEB row is a three-seed mean. Candidate generation, filtering, training, and frame selection differ, and published tail results do not supply this study's fixed tail membership.

Table 14: Published references and CSEB detector f-mAP (%); protocols differ.

Model	Agent	Action	Loc.	Duplex	Triplet
3D-RetinaNet (benchmark)	15.57	12.30	12.40	10.50	6.20
SlowFast	17.45	15.35	14.42	9.66	5.51
MViT	20.75	13.10	15.41	10.17	6.63
MViTv2	25.72	15.31	19.26	10.92	7.06
MeMViT	29.27	18.84	26.11	12.09	7.28
STMixer	24.70	17.12	20.98	15.16	9.33
CycleACR (SlowFast)	33.86	25.23	32.80	17.17	10.06
FRCB	33.64	23.87	32.33	17.84	11.71
CSEB (ours, mean)	35.25	24.75	29.64	24.36	15.75

The CSEB mean is 6.52 percentage points above the published FRCB best-result duplex score and 4.04 points above its triplet score. CycleACR's published action and location results remain higher by 0.48 and 3.16 points. FRCB also reports a five-run triplet mean of $11.53 \pm 0.35\%$

in Table 5, distinct from the 11.71% best-result row. These numerical differences provide context, but do not establish a matched state-of-the-art result or a measured tail advantage over FRCB.

4.8 Measured Examples of Composition Ranking

The worked calculation in Section 3.16 explains the operations using synthetic scores. The examples here instead use recorded detector outputs from the original-encoder crop study, seed 0. They were selected after inspecting results to illustrate contrasting behavior; they are not a random sample, a success rate, or evidence that the final contextual system resolves these cases.

Figure 26 shows a large vehicle annotated as stopped at a junction in ROAD-Waymo validation video train_00797, frame 18. The candidate overlaps the ground-truth box at IoU 0.679, above the matching threshold. This triplet has 1,895 training boxes and belongs to the deeper-tail group. Stage 4 ranks a successful match at 288, Stage 5 at 81,918, and Stage 6 at 4,708. With the diagnostic top-615 prediction budget, Stage 4 retains the match while the two composition models do not. The budget equals the evaluation ground-truth count for this class; it is an analysis convention, not a deployment threshold.

For the same recorded candidate, the confidence-weighted LarVeh-Stop-Jun scores are 0.731, 0.071, and 0.309 for Stages 4, 5, and 6. These scores and the successful-match ranks describe different aspects of the trace: the score is for the same chosen box, whereas the first successful match can involve another candidate for that object. Stage 6 improves this instance relative to Stage 5, but does not recover Stage 4's ranking. It would be incorrect to describe this as a complete localization failure or to infer a rescue by the later CSEB model.

A contrasting Bus-Stop-VehLane case in validation video train_00635, frame 186, has candidate IoU 0.965. Successful-match ranks are 101, 2,878, and 325 for Stages 4–6. Under its top-583 diagnostic budget, Stage 6 restores an included match relative to Stage 5, while Stage 4 ranks it

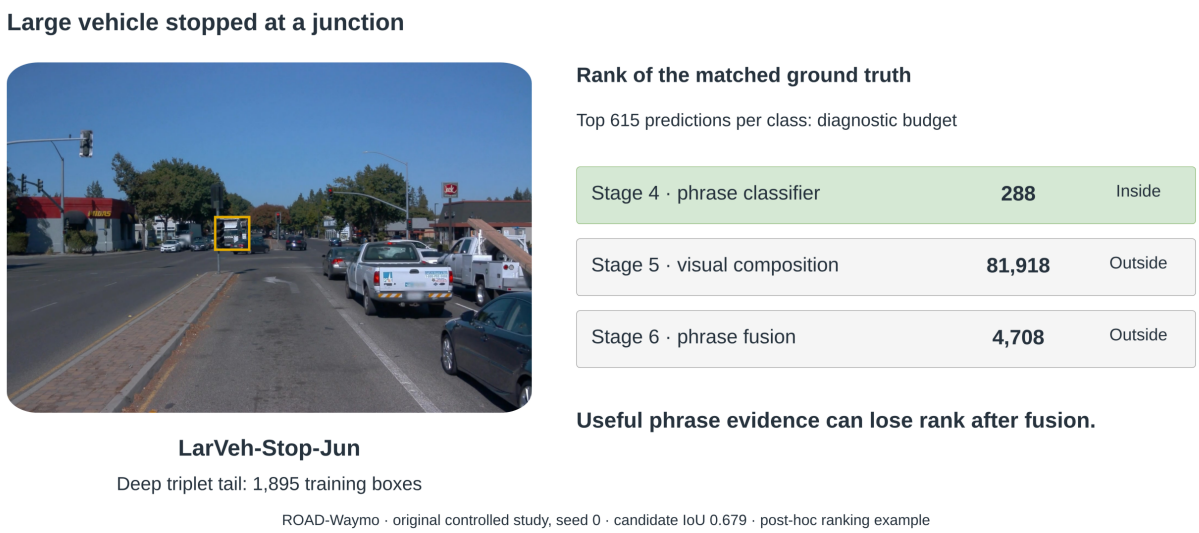


Figure 26: A measured deep-tail ranking example on ROAD-Waymo [2]. This original-encoder seed-0 diagnostic illustrates loss of useful phrase ranking after learned composition. The still illustrates the annotated event; temporal evidence is needed to assess motion.

earlier. These selected examples illustrate heterogeneous effects; aggregate AP remains necessary to assess the full population.

4.9 Illogical Outputs and Ordinary Errors

A detected event should tell a coherent story about one agent at one time. However, restricting outputs to the benchmark's valid composition labels does not guarantee agreement between independently scored primitive and composition labels. For example, Bus-Stop-VehLane is a meaningful event label, but emitting it for a candidate whose selected agent label is Car gives an inconsistent description of that candidate. The present pipeline does not enforce an agent-class mask on composition scores. Neither composition learning nor language attention alone guarantees logical consistency.

The examples in Table 15 are deliberately constructed teaching cases, not recorded outputs or measured violation rates. Let T denote Bus-Stop-VehLane and let B, S, and V denote its Bus,

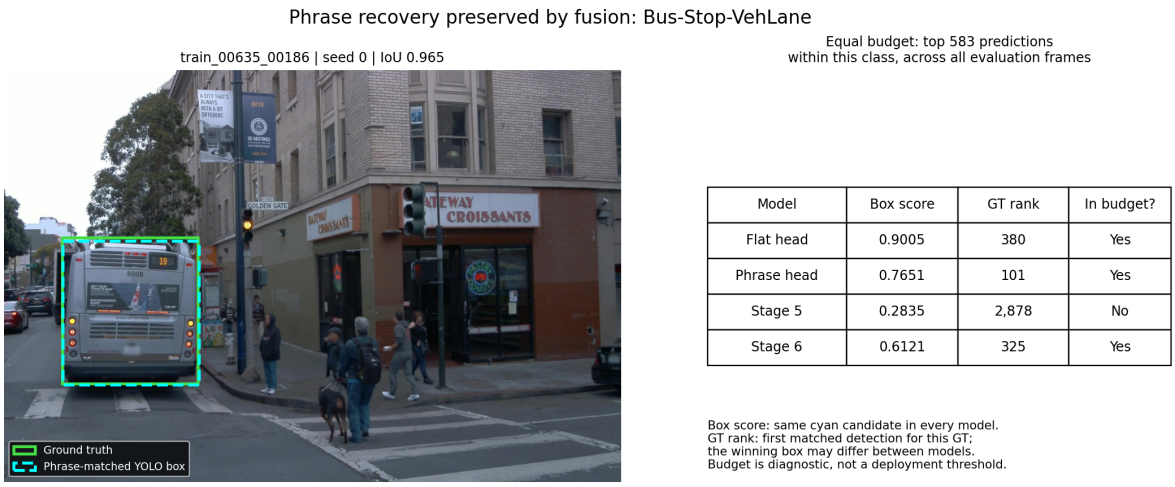


Figure 27: A measured Bus-Stop-VehLane example on ROAD-Waymo [2], original-encoder study, seed 0. "Phrase head" denotes Stage 4; "Flat head" is the separate no-composition baseline. Green marks ground truth and cyan the selected candidate. The same-candidate scores and first successful-match ranks are different diagnostics.

Stop, and VehLane constituents for the same candidate and time. A coherent hard-label interpretation requires

$$T \Rightarrow (B \wedge S \wedge V). \tag{4.2}$$

For illustration only, define $z_k = \mathbb{1}[s_k \geq \tau]$ with $\tau = 0.5$, using scores s_k on a common emitted-score scale. Then an implication violation is

$$I_T = z_T \left(1 - z_B z_S z_V\right). \tag{4.3}$$

This threshold illustrates a logical test; it is not the AP evaluator, an adopted deployment rule, or a claim that these scores are calibrated probabilities.

The missing-action example can be read step by step: the threshold selects the triplet ($z_T = 1$), Bus ($z_B = 1$), and VehLane ($z_V = 1$), but not Stop ($z_S = 0$). Equation 4.3 therefore gives

Table 15: Illustrative logical conflicts and control cases. All scores are synthetic; each row concerns one candidate at one time.

Case	Example output	Interpretation
Agent conflict	Selected agent: Car. $s_T = 0.80$, $s_B = 0.10$.	The event says Bus while the selected agent says Car. The triplet is valid in the vocabulary, but the description disagrees across heads.
Missing action	$s_T = 0.80$, $s_B = 0.90$, $s_S = 0.15$, $s_V = 0.85$.	The event is selected while Stop is omitted: $I_T = 1$. The triplet does not imply that the action head actually selected Stop.
Missing location	$s_T = 0.80$, $s_B = 0.90$, $s_S = 0.85$, $s_V = 0.10$.	The event asserts VehLane while its location constituent is omitted: $I_T = 1$.
Coherent but wrong	$s_T = 0.80$, $s_B = s_S = s_V = 0.90$, but the annotated agent is Car.	$I_T = 0$: the outputs agree with one another but disagree with ground truth. Logical consistency is not recognition accuracy.

$1(1 - 1 \cdot 0 \cdot 1) = 1$. Selecting Stop as well would remove this particular conflict, but could introduce a false-positive action. Suppressing the triplet could instead remove a correct detection. These tradeoffs explain why a consistency repair must be evaluated for recognition accuracy as well as agreement; neither repair is implemented or credited with improvements here.

Not every unusual combination is illogical. ROAD-Waymo permits multiple action and location labels. The measured large-vehicle example in Figure 26 has both LarVeh-Stop-Jun and LarVeh-Stop-IncomLane annotations: junction and incoming-lane descriptions can apply together. Rejecting all multiple-location predictions would therefore discard valid labels. Similarly, the low ranking of a true triplet in that example is a recognition failure, not by itself evidence of a logical contradiction. A still image also cannot establish a contradictory motion prediction without the temporal clip and the label definitions.

A measured consistency audit remains future work. It would retain full primitive and composition vectors for the same candidate and time, declare its decision rule, decode constituents from the verified composition-label strings, and report conflict counts with their denominator alongside detector AP. Ground-truth review would distinguish valid multi-label cases from genuine conflicts. The current selected-case traces establish ranking behavior, not a final-CSEB violation rate. No claim is made that CSEB eliminates illogical outputs.

4.10 Interpretation and Validity

The strongest component comparisons hold the surrounding pipeline fixed: Stage 2-to-3 changes composition outputs, Stage 5-to-6 adds phrase-composition inputs, and the contextual studies compare losses and expert predictions within one family. Crop-to-context and encoder-scaling comparisons change several components and support system-level conclusions.

All results use a validation population that informed subsequent research choices. Internal training partitions select contextual checkpoints and blend weights, but this does not make the final validation set independent of development. Stages 0–3 are historical single runs; later three-seed means measure optimization variability on fixed data. Independent test-set performance, tube-level detection, unseen-class generalization, and deployment safety remain untested.

CHAPTER 5: Conclusion and Future Work

This thesis investigated long-tail compositional road-event detection using a decoupled detector and event classifier. The completed study extends from a replicated joint detector through direct regional classification, learned composition, semantic crop classification, and contextual expert blending. The results show that improved agent localization alone does not resolve event recognition: early score transfer improves agent detection while reducing triplet mAP. Direct feature access and learned composition provide a stronger basis for subsequent semantic and contextual models.

5.1 Principal Findings

Learned composition improves the controlled Stage 2-to-3 comparison from 6.65% to 9.73% triplet mAP while keeping candidate boxes and primitive outputs fixed. Semantic crop classification reaches 5.77% tail mAP, but its 9.07% overall triplet result remains below Stage 3. Adding phrase-composition evidence to the crop composition MLP changes triplet mAP only from 11.23% to 11.28%, with a deeper-tail regression. These findings motivate evaluating overall, common, and rare-class behavior separately.

The contextual family combines crop features, actor-region evidence, the wider scene, box geometry, and phrase representations. Original-encoder language controls favor the real phrase bank on tail means, while their small sample and geometry differences limit a purely semantic interpretation. Classification-only contextual heads already use language; the additional contrastive objective makes small, mixed changes. Its value in the final model includes complementarity with a separately trained classification expert.

The final Contextual Semantic Expert Blend achieves $15.75 \pm 0.27\%$ triplet mAP and $9.00 \pm 0.64\%$ tail mAP across three seeds, where the uncertainty term is sample standard deviation.

Relative to the original-encoder DCB blend, the mean increases are 2.32 percentage points overall and 2.77 points on the tail. Approximately 65% of the overall improvement comes arithmetically from the 47 tail classes, and common-class performance also improves. This result supports the final system under the fixed validation protocol; it does not isolate encoder size, language, or loss as the sole cause.

5.2 Limitations

The evaluation population is a shared subset of ROAD-Waymo validation used during development. Checkpoint selection and blend fitting use held-out training videos, but subsequent research decisions were informed by validation results. Three repeated seeds measure optimization sensitivity on this fixed population and do not establish environmental generalization. Published comparisons differ in candidates, filtering, training, and repetition counts, so higher reported mAP does not constitute a matched state-of-the-art test.

The scaling comparison changes both pretrained towers, the text projection, and cache precision. Language-bank controls reported here concern the original-encoder focal scene-attention head. They cannot establish the same effect in the scaled DCB blend. The fixed class vocabulary does not test unseen combinations, and valid composition columns do not guarantee agreement with independently scored primitives. The study also does not establish end-to-end real-time performance or driving safety.

5.3 Future Work

Matched language-bank controls in the scaled CSEB, followed by separate visual- and text-checkpoint comparisons, would narrow the attribution of the final gains. These comparisons should hold cache precision and head interfaces fixed. Additional seeds and an independent evaluation population would strengthen reproducibility and generalization claims.

Per-class analysis should identify persistent rare-composition errors and distinguish recognition failures from missing or inaccurate boxes. Causal temporal windows, cross-domain evaluation, and full-pipeline latency measurements are needed before online use. Structured output constraints could test consistency separately from AP.

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